

Spin–Momentum Impedance and Filtering by a Spin-Coupled Absorbing Boundary Condition

Alireza Jozani*

Departments of Physics and Mathematics, Eberhard-Karls-Universität Tübingen
Auf der Morgenstelle 10, 72076 Tübingen, Germany

June 24, 2026

Abstract

Absorbing boundaries are often treated as scalar sinks. Here we show that a spin-coupled absorbing boundary for a Pauli particle acts instead as a spin–momentum impedance. Its tangential boundary symbol has two branches, $i\kappa \pm |\boldsymbol{\xi}|$, coupling normal absorption to in-plane momentum. In a harmonic guide, the transverse ground state samples $|\boldsymbol{\xi}| \sim \ell_{\perp}^{-1} \sim \sqrt{\omega}$; narrowing the guide therefore strengthens a local evanescent boundary response without introducing a bulk potential barrier. Solving the detector-present spinor absorbing-boundary evolution, we identify boundary-induced filtering: the prompt detector flux is suppressed, the fixed-window detected fraction is reduced, and a delayed oscillatory sector appears. Over that window the restricted mean detection time is fitted by $A + B\sqrt{\omega}$, with setup-dependent coefficients. The robust result is a spin–momentum filtering mechanism with boundary scale $|\boldsymbol{\xi}| \sim \sqrt{\omega}$, not a universal arrival-time law.

1 Introduction

Predicting when a quantum detector clicks is not fixed by the usual Born rule for position at a prescribed time. This arrival- and detection-time problem has led to several inequivalent proposals, including flux-based rules, detector-free trajectory arrival times, complex absorbing potentials, and absorbing boundary conditions [1–6]. A central point is that a detector-free arrival time and a detector-present click time are generally different physical objects: the detector is not a passive marker of a pre-existing crossing event, but part of the dynamical boundary problem that defines the click-time distribution.

Absorbing boundaries are often introduced to remove probability from an open quantum evolution. In detector models this loss is physical: the squared norm $\|\Psi_t\|^2$ is the probability that no click has yet occurred, and its loss rate is the click-time density [2, 5, 6]. The boundary is therefore not merely a surface where probability disappears. Like an impedance in wave physics, it is a boundary law determining how incident components are absorbed and how the undetected components are reflected, mixed, or filtered within the no-click evolution. In numerical settings such interfaces are often introduced as absorbing or radiation devices [7, 8]; in the detector model considered here, the same response is part of the measurement, not merely a numerical device [2, 5, 6].

In this work we study a spin-coupled absorbing boundary condition, or spinor ABC, whose impedance is matrix-valued. For a nonrelativistic spin-1/2 particle in a harmonic waveguide, the absorbing boundary couples the normal derivative of the spinor to its tangential derivatives.

*jozani.alireza@gmail.com; alireza.jozani@uni-tuebingen.de

Tangential Fourier decomposition at the detecting surface turns the boundary condition into a 2×2 matrix relation $\partial_z \hat{\Psi} = \mathcal{C}(\xi) \hat{\Psi}$. The eigenvalues $i\kappa \pm |\xi|$ of $\mathcal{C}(\xi)$ define two spin-momentum branches. For the harmonic transverse ground state, $\ell_\perp = \omega^{-1/2}$ in dimensionless units; hence increasing ω narrows the guide and raises the typical tangential scale sampled at the boundary, so that $|\xi| \sim \ell_\perp^{-1} \sim \sqrt{\omega}$. This local boundary scale underlies the delayed detector-present roof-flux response studied below.

At first sight this confinement dependence is surprising, since the bulk Hamiltonian used here is spin independent and separable, so changing ω changes the normalized transverse profile but does not by itself change the plane-integrated first-pass longitudinal Pauli flux. Thus the square-root scale observed in the detector response is not generated by ordinary first-pass bulk propagation. It must enter when the wave interacts with the spinor ABC.

This response is detector-present. The observable is the roof flux, equivalently the norm loss, of a specified nonunitary absorbing-boundary evolution. It is not the no-detector Bohmian first-arrival distribution of a freely evolving Pauli wave [3, 4, 9, 10]. Nor is the spin-coupled absorbing boundary assumed to describe all possible detecting screens; it is a particular idealized hard-detector model, a restriction emphasized by recent scattering-theory comparisons of absorbing detectors [11].

The Das-Dürr waveguide geometry provides a sharp comparison between detector-free arrival and detector-present detection [4]. In the same separable guide, complex absorbing potentials and the spin-decoupled absorbing boundary behave as scalar absorbers: they may generate reflected tails through detector back-action, but they do not read transverse confinement in this way [12, 13]. By contrast, the spin-coupled absorbing boundary suppresses the prompt roof-flux peak, lowers the detected fraction in a fixed observation window, and converts the transverse confinement into a delayed oscillatory sector, while showing no appreciable dependence on the initial Bloch angles in the tested regimes [13]. Those simulations left the confinement dependence unexplained. Here we identify its local origin, derive the two-branch boundary symbol $i\kappa \pm |\xi|$, and show how this boundary scale organizes the observed ω -dependent detector-response signature.

2 Model and detector observable

We work in units $\hbar = m = 1$. The bulk Hamiltonian is spin independent,

$$H = -\frac{1}{2}\Delta + \frac{1}{2}\omega^2[x^2 + y^2], \quad (1)$$

and the initial state is factorized as

$$\Psi_0(x, y, z) = \chi_\omega(x, y) \phi_0(z) \eta, \quad (2)$$

where $\eta \in \mathbb{C}^2$ is a constant spinor and χ_ω is the transverse harmonic ground state. For the finite-window confinement study below, ϕ_0 is a right-moving Gaussian longitudinal packet centered midway along the guide. The finite guide has a reflecting lower end at $z = 0$, and the detecting surface is the roof

$$\Sigma_L = \{z = L\}. \quad (3)$$

On this surface we impose the spin-coupled absorbing boundary condition

$$(\boldsymbol{\sigma} \cdot \boldsymbol{\nabla})\Psi = i\kappa\sigma_z\Psi, \quad \kappa > 0. \quad (4)$$

This is the nonrelativistic limit of the semi-ideal Dirac absorbing boundary [14]; well-posedness and contraction-semigroup formulations are discussed in Refs. [15, 16]. For the Pauli current

$$\mathbf{j}^P = \text{Im}(\Psi^\dagger \boldsymbol{\nabla} \Psi) + \frac{1}{2} \boldsymbol{\nabla} \times (\Psi^\dagger \boldsymbol{\sigma} \Psi), \quad (5)$$

the spinor ABC gives the outward roof flux

$$j_z^P(x, y, L, t) = \kappa \rho(x, y, L, t), \quad \rho = \Psi^\dagger \Psi. \quad (6)$$

By the continuity equation, with no flux through the remaining faces, the detector-present click-time density is

$$g(t; \omega) = \kappa \int_{\Sigma_L} \rho(x, y, L, t) dx dy = -\frac{d}{dt} \|\Psi_t\|^2. \quad (7)$$

For a finite observation horizon T , define the detected fraction

$$D_T(\omega) = \int_0^T g(t; \omega) dt, \quad (8)$$

and the finite-window restricted mean detection time,

$$\mu^*(T; \omega) = \int_0^T S(t; \omega) dt, \quad S(t; \omega) = \|\Psi_t\|^2, \quad (9)$$

or,

$$\mu^*(T; \omega) = \mathbb{E}[\min(\tau, T)]. \quad (10)$$

Here τ is the random detector click-time. We use this restricted mean detection time because, at strong confinement, a substantial part of the probability remains undetected at the end of the finite observation window. Bohmian histograms shown in the plots are only Monte Carlo samples of the same detector-present flux law. These trajectories obey

$$\dot{\mathbf{Q}}(t) = \frac{j^P[\Psi_t^{\text{ABC}}](\mathbf{Q}(t))}{\rho[\Psi_t^{\text{ABC}}](\mathbf{Q}(t))}. \quad (11)$$

Here $\mathbf{Q}(t)$ denotes a sampled Pauli-current trajectory and Ψ_t^{ABC} denotes the nonunitary spinor-ABC evolution.

We now present the confinement-dependent roof flux and then derive the boundary-symbol mechanism responsible for the $\sqrt{\omega}$ scale.

3 Confinement-induced spin-momentum filtering by the spin-coupled ABC

Figure 1 shows the central numerical observation. The red curve in Fig. 1 is the detector-free one-dimensional Gaussian current through $z = L$ for the same longitudinal packet, with the transverse factor integrated out. It is not a detector model or fit input; it only marks the free crossing scale defined in detail in Ref. [13]. Only the transverse confinement parameter ω is varied; the longitudinal packet, detector parameter, initial spinor, and observation window are fixed. Increasing ω narrows the transverse ground state and strongly suppresses the prompt roof-flux peak. This prompt suppression is the most direct finite-window signature of the boundary response. The detected fraction drops from $D_{20} \simeq 0.93$ at $\omega = 1$ to $D_{20} \simeq 0.42$ at $\omega = 300$. Figure 1(c) summarizes the same confinement sweep: over the sampled ω -values, the restricted mean detection time is fitted by

$$\mu^*(20; \omega) \simeq 4.084 + 0.638\sqrt{\omega}. \quad (12)$$

The fit in Eq. (12) is a finite-window observation, not an asymptotic law. Its coefficients would depend on the prepared wave packet, detector parameter, guide geometry, and observation time. The robust feature is the square-root scale. We now identify its boundary origin and the finite-guide mechanism that makes this scale visible in the roof-flux detection time statistic.

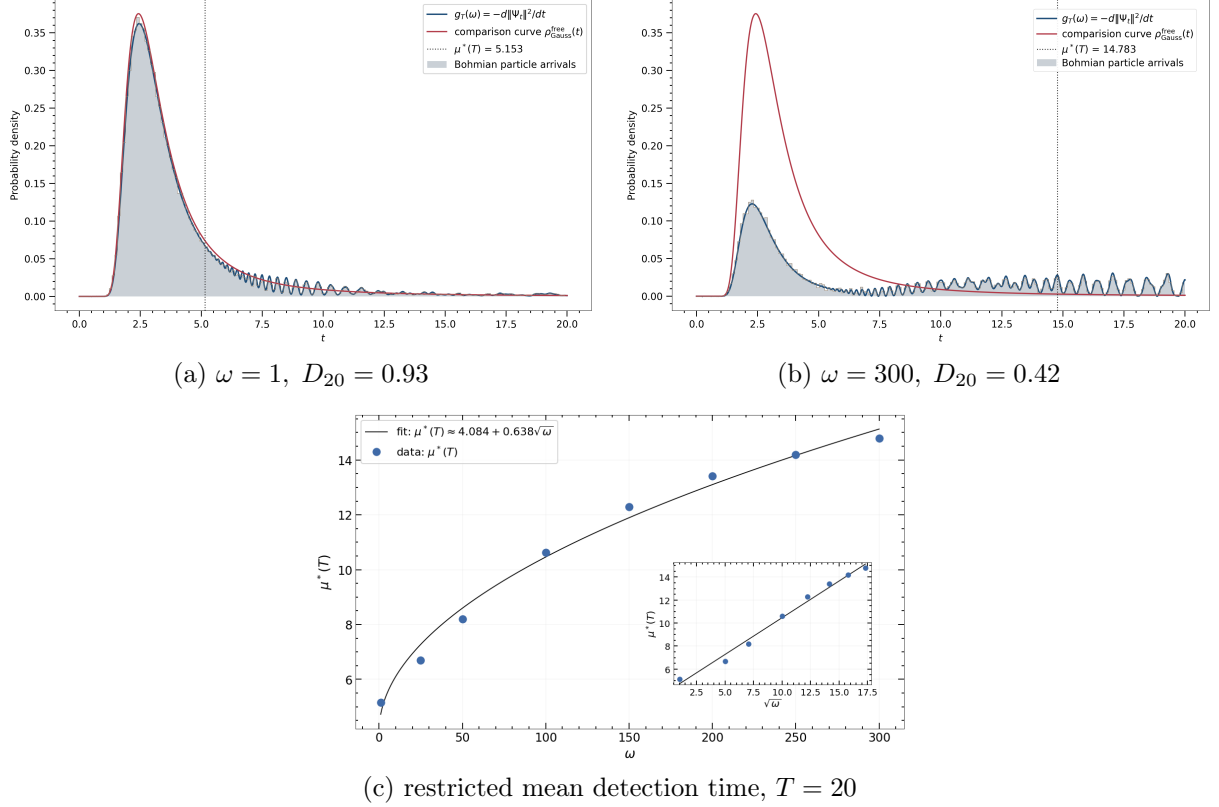


Figure 1: Confinement-dependent roof-flux response of the spin-coupled ABC. (a),(b) Detector-present roof-flux density $g(t; \omega) = -d\|\Psi_t\|^2/dt$ for $\omega = 1$ and $\omega = 300$. red: detector-free Gaussian comparison current; blue: spinor-ABC roof flux; gray histograms: trajectory samples of the same spinor-ABC roof-flux law, not detector-free Bohmian arrival times. Only ω is varied; $\kappa = k_z = \pi$, $\sigma_{\parallel} = 0.5$, and $T = 20$ are fixed. Increasing confinement suppresses the prompt peak and lowers D_{20} from 0.93 to 0.42. (c) Restricted mean detection time $\mu^*(T; \omega)$, fitted by $\mu^*(20; \omega) \simeq 4.084 + 0.638\sqrt{\omega}$ for several ω . Details of the numerical setup, full confinement sweep, finite-window bookkeeping, and convergence tests are given in the Supplementary Information [17], Secs. S2–S4.

4 Boundary-symbol mechanism and finite-guide memory

We discuss the mechanism through three logical steps: (i) ordinary first-pass bulk propagation is excluded as the source of the ω -dependence; (ii) the spinor ABC is shown to introduce the local roof scale $R = |\boldsymbol{\xi}|$; and (iii) finite-guide memory provides a natural route by which undetected, branch-reweighted components can return to the roof and contribute to the delayed detection-time sector in Fig. 1.

(i) The square-root scale is not generated by ordinary first-pass bulk propagation. Before significant feedback from the detecting roof, the spin-independent separable bulk evolution has the first-pass form

$$\Psi_{\text{fp}}(x, y, z, t) = \chi_\omega(x, y, t) \phi_{\text{fp}}(z, t) \eta, \quad (13)$$

where $\phi_{\text{fp}}(z, 0) = \phi_0(z)$. The transverse harmonic ground state changes only by an overall phase, so its density is unchanged, and the spinor η remains constant because the bulk Hamiltonian is spin independent. For this first-pass field, the Pauli current can then be written

$$\mathbf{j}_{\text{fp}}^P = \mathbf{j}_{\text{fp}}^{\text{conv}} + \frac{1}{2} \nabla \times (\rho_{\text{fp}} \mathbf{s}_\eta), \quad \mathbf{s}_\eta := \eta^\dagger \boldsymbol{\sigma} \eta, \quad \rho_{\text{fp}} := |\Psi_{\text{fp}}|^2. \quad (14)$$

Since $\mathbf{s}_\eta$ is constant in the first-pass separable approximation, the z -component of the spin-curl term is a transverse divergence. By Stokes' theorem on the transverse cross-section, together with the lateral Dirichlet conditions, its cross-section integral vanishes. Hence, for an interior plane Σ_b below the roof,

$$\int_{\Sigma_b} j_{\text{fp},z}^P dA = \int_{\Sigma_b} \text{Im}(\Psi_{\text{fp}}^\dagger \partial_z \Psi_{\text{fp}}) dA = \text{Im}(\phi_{\text{fp}}^* \partial_z \phi_{\text{fp}})(z_b, t). \quad (15)$$

The leading plane-integrated first-pass flux is therefore independent of the transverse confinement parameter ω . More details on the first-pass bulk calculation are given in the Supplementary Information [17], Sec. S5.

(ii) The observed confinement dependence therefore must enter when the wave interacts with the detecting boundary. The spinor ABC introduces this scale through the tangential symbol $\mathcal{C}(\boldsymbol{\xi})$ of the boundary operator. The same roof condition, Eq. (4), can be written componentwise as

$$\begin{aligned} \partial_z \psi_\uparrow &= i\kappa \psi_\uparrow - (\partial_x - i\partial_y) \psi_\downarrow, \\ \partial_z \psi_\downarrow &= i\kappa \psi_\downarrow + (\partial_x + i\partial_y) \psi_\uparrow. \end{aligned} \quad (16)$$

Equivalently,

$$\partial_z \Psi = \mathcal{C} \Psi, \quad \mathcal{C} = \begin{pmatrix} i\kappa & -D_- \\ D_+ & i\kappa \end{pmatrix}, \quad D_\pm = \partial_x \pm i\partial_y. \quad (17)$$

Therefore the detector couples normal absorption to tangential derivatives. For a tangential Fourier mode $\boldsymbol{\xi} = (\xi_x, \xi_y)$, Eq. (17) gives

$$\mathcal{C}(\boldsymbol{\xi}) = i\kappa I + (\hat{z} \times \boldsymbol{\xi}) \cdot \boldsymbol{\sigma}. \quad (18)$$

Writing

$$R := |\boldsymbol{\xi}|, \quad \Gamma(\boldsymbol{\xi}) = \frac{(\hat{z} \times \boldsymbol{\xi}) \cdot \boldsymbol{\sigma}}{R}, \quad \Gamma^2 = I. \quad (19)$$

Thus

$$\mathcal{C}(\boldsymbol{\xi}) = i\kappa I + R\Gamma(\boldsymbol{\xi}). \quad (20)$$

The two tangential spin-momentum projectors are then

$$\Pi_\pm(\boldsymbol{\xi}) = \frac{1}{2}(I \pm \Gamma(\boldsymbol{\xi})), \quad (21)$$

and the corresponding boundary eigenbranches are

$$\lambda_{\pm}(\boldsymbol{\xi}) = i\kappa \pm R. \quad (22)$$

This is the key point: the detector imposes not only the impedance scale κ , but also the tangential spin-momentum scale $R = |\boldsymbol{\xi}|$. The same branch structure gives the local normal response. Let $W(\boldsymbol{\xi}, t) = \widehat{\Psi}(\boldsymbol{\xi}, L, t)$ be the roof trace. For a small auxiliary inward depth $\epsilon = L - z$, the homogeneous normal continuation generated by the boundary relation is

$$\widehat{\Psi}(\boldsymbol{\xi}, L - \epsilon, t) = e^{-i\kappa\epsilon} B_{\text{br}}(\epsilon, \boldsymbol{\xi}) W(\boldsymbol{\xi}, t) + \widehat{\mathcal{R}}_{\text{Duh}}, \quad (23)$$

where

$$B_{\text{br}}(\epsilon, \boldsymbol{\xi}) W = e^{-R\epsilon} \Pi_+ W + e^{R\epsilon} \Pi_- W. \quad (24)$$

This is a matrix-valued branch filter acting on the spinor amplitude $W(\boldsymbol{\xi}, t)$: it decomposes the roof trace into the two tangential spin-momentum components $\Pi_{\pm} W$ and assigns them the auxiliary normal weights $e^{-R\epsilon}$ and $e^{R\epsilon}$. The term $\widehat{\mathcal{R}}_{\text{Duh}}$ is a Duhamel remainder measuring the deviation of the true time dependent Schrödinger equation (TDSE) solution from this homogeneous boundary continuation away from the roof; it has no zeroth- or first-order contribution at $\epsilon = 0$, as shown in the Supplementary Information [17], Sec. S7. All physical detection statistics remain defined at the roof, $\epsilon = 0$. Therefore B_{br} is not a physical propagation law through a detector layer, but a local boundary-symbol probe

$$B_{\text{br}}(0, \boldsymbol{\xi}) = I, \quad \partial_{\epsilon} B_{\text{br}}(\epsilon, \boldsymbol{\xi})|_{\epsilon=0} = -R\Gamma(\boldsymbol{\xi}). \quad (25)$$

The branch structure is not a direct multiplicative suppression of the roof density; it enters the first inward normal response. Therefore the same tangential radius $R = |\boldsymbol{\xi}|$ that labels the roof mode controls the local normal boundary response. For the harmonic transverse ground state, $\ell_{\perp} = \omega^{-1/2}$, so the typical tangential Fourier scale is set by

$$R = |\boldsymbol{\xi}| \sim \ell_{\perp}^{-1} \sim \sqrt{\omega}. \quad (26)$$

This is the local boundary scale. To express how it enters a normalized first time moment, define an auxiliary boundary layer response measure

$$d\mu_{\omega, \epsilon}^{\text{bl}}(\boldsymbol{\xi}, t) := |B_{\text{br}}(\epsilon, \boldsymbol{\xi}) W(\boldsymbol{\xi}, t)|^2 d^2\xi dt. \quad (27)$$

This is not a click-time distribution at $z = L - \epsilon$: the variable t is only the time label of the roof trace $W(\boldsymbol{\xi}, t)$, and the physical detector observable remains the roof flux at $z = L$. Since the scalar phase drops out of densities and the Duhamel term has no first-order contribution at the roof, the first variation is computed from the homogeneous branch probe. With $B_{\text{br}} = e^{-\epsilon R\Gamma}$,

$$|B_{\text{br}} W|^2 = W^{\dagger} e^{-2\epsilon R\Gamma} W = |W|^2 b_{\omega, \epsilon}. \quad (28)$$

where

$$b_{\omega, 0} = 1, \quad \partial_{\epsilon} b_{\omega, \epsilon}|_{\epsilon=0} = Ra_{\omega}, \quad (29)$$

and

$$a_{\omega}(\boldsymbol{\xi}, t) := -2 \frac{W^{\dagger} \Gamma(\boldsymbol{\xi}) W}{W^{\dagger} W}, \quad |a_{\omega}| \leq 2. \quad (30)$$

To normalize the auxiliary boundary-response measure, let $d\nu_{\omega}^{(0)} = d\mu_{\omega, 0}^{\text{bl}} / \int d\mu_{\omega, 0}^{\text{bl}}$. After normalization,

$$\mathbb{E}_{\nu_{\omega, \epsilon}^{\text{bl}}} [F] = \frac{\mathbb{E}_{\nu_{\omega}^{(0)}} [F b_{\omega, \epsilon}]}{\mathbb{E}_{\nu_{\omega}^{(0)}} [b_{\omega, \epsilon}]}. \quad (31)$$

Differentiating at $\epsilon = 0$ gives

$$\partial_\epsilon \mathbb{E}_{\nu_{\omega,\epsilon}^{\text{bl}}}[F]|_{\epsilon=0} = \text{Cov}_{\nu_{\omega}^{(0)}}(F, Ra_\omega). \quad (32)$$

Take $F = t$, the local first-order boundary sensitivity is

$$\Lambda_\omega = \text{Cov}_{\nu_{\omega}^{(0)}}(t, Ra_\omega). \quad (33)$$

For the harmonic transverse family, $R = \sqrt{\omega}s$, with s dimensionless and order one. Hence

$$\Lambda_\omega = \sqrt{\omega} \text{Cov}_{\nu_{\omega}^{(0)}}(t, sa_\omega) =: \sqrt{\omega} \beta_\omega. \quad (34)$$

Thus Λ_ω is a local first-order boundary sensitivity, not the observed finite-window delay itself. Equation (34) shows that the first boundary-response contribution factorizes into the explicit transverse scale $\sqrt{\omega}$ and a dimensionless roof-trace covariance β_ω . The latter is not universal: it depends on the detector-present roof trace and may change with the prepared wave packet, detector parameter, guide geometry, and observation window. The finite-grid diagnostics in the Supplementary Information [17], Sec. S7, show $\beta_\omega = O(1)$ over the sweep, with no competing power-law growth. Thus the local calculation identifies the $\sqrt{\omega}$ scale, not the setup-dependent value or sign of B in the finite-window fit $A + B\sqrt{\omega}$.

(iii) In the finite guide, the local boundary response becomes visible in the roof-flux statistic through finite-guide memory. During the first near-roof encounter, the spinor ABC does not remove a scalar fraction of the incident packet. The amplitude that is not absorbed has already sampled the matrix boundary symbol, has been attenuated by detector loss, and has been reweighted between the two tangential spin-momentum branches. The surviving no-click amplitude that subsequently propagates back into the guide is therefore not a scalar reflected copy of the incident longitudinal packet.

This distinction is important for interpreting the delayed oscillatory sector. No detected probability is returning to the detector. The returning object is the undetected amplitude after it has been filtered by the spinor ABC. Since this component inherits the local boundary scale $R = |\xi|$, and since the harmonic transverse family samples $R \sim \ell_\perp^{-1} \sim \sqrt{\omega}$, finite-guide memory converts the local branch reweighting into a delayed roof-flux contribution. The numerical coefficients of a finite-window fit such as $A + B\sqrt{\omega}$ remain dependent on the source packet, detector parameter, guide length, and observation window. The robust point is that the spinor boundary makes the detector-present roof flux sensitive to the tangential momentum scale. A reduced flux-level bookkeeping formulation of this memory effect is given in the Supplementary Information, [17] Sec. S8; it is not an exact mode-resolved TDSE decomposition and is not used as a fit model.

5 Conclusion

We have shown that the spinor ABC is not a scalar absorbing sink but a local spin-momentum impedance. At the detecting roof, each tangential mode decomposes into two spin-momentum components $\Pi_\pm W$, with boundary eigenbranches $i\kappa \pm |\xi|$. In a harmonic guide the roof trace samples $R = |\xi| \sim \ell_\perp^{-1} \sim \sqrt{\omega}$, so increasing transverse confinement directly changes the local detector-present boundary response. Equation (24) gives the local meaning of this filtering: after the scalar phase is removed, the auxiliary inward normal response weights the two spin-momentum components by $e^{\mp R\epsilon}$. The $e^{-R\epsilon}\Pi_+$ factor is an evanescent normal factor generated by the boundary symbol itself, not by propagation through a classically forbidden bulk region or a potential barrier. The complementary $e^{R\epsilon}\Pi_-$ factor is the opposite branch of the same auxiliary continuation, not a gain channel. The physical detection statistic remains the roof flux at $z = L$.

Since the spin-independent separable bulk first-pass flux is independent of ω , the observed confinement dependence is generated at the detecting boundary rather than by ordinary bulk propagation. Together with finite-guide memory of the undetected, branch-reweighted amplitude, the local scale $R \sim \sqrt{\omega}$ organizes the numerical signatures in Fig. 1: suppression of the prompt roof flux, reduction of D_{20} , and growth of the finite-window restricted mean.

Accordingly, the fit

$$\mu^*(20; \omega) \simeq 4.084 + 0.638\sqrt{\omega} \quad (35)$$

should be read only as a finite-window diagnostic for the specified prepared wave packet, detector parameter, guide geometry, and observation time. The constants are effective and setup dependent. The robust observable signature is the confinement-controlled deformation of the detector-present roof-flux distribution by a matrix-valued absorbing boundary, not a universal arrival-time law.

A direct physical realization of this response would require engineering an absorbing interface whose effective entrance impedance realizes the tangential branches $i\kappa \pm |\xi|$. The controls in the Supplemental Material support this specificity in two complementary ways. First, when the spinor ABC is retained, representative bulk spin perturbations, including Zeeman-type and spin-orbit terms, do not substantially remove the delayed sector. Second, when the spinor boundary is replaced by generic spin-dependent absorbing Hamiltonian layers, the spinor-ABC confinement trend is not robustly reproduced. The inverse-engineered layer approaches the response only when the branch impedances $i\kappa \pm |\xi|$ are built in by construction; detailed controls, parameter scans, and physical scale estimates are given in the Supplementary Information [17], Secs. S9–S11.

The broader implication is that absorbing interfaces with internal degrees of freedom need not behave as passive scalar probability sinks. Even when the bulk Hamiltonian is spin independent and separable, the detector boundary can read tangential momentum through its local symbol and imprint that scale on click-time statistics. Thus detector-present click-time statistics can contain information about the boundary impedance, not only about the freely propagated incident packet.

Declarations and acknowledgments

Data and code availability. Processed data underlying Fig. 1 and the Supplementary figures, representative density animations, and the Python scripts for the Crank–Nicolson/GMRES simulations, diagnostics, post-processing, and figure generation are available at <https://github.com/jloOop/Boundary-Spin-Momentum-Filtering-Data>. The repository includes parameters, reduced data, and reproducibility notes needed to reproduce the reported figures. Larger raw HPC outputs are omitted because of size; selected additional outputs are available from the author upon reasonable request. The solver follows the implementation described in Appendix A of Ref. [13].

Competing interests. The author declares no competing interests.

Author contributions. A.J. is the sole author of this work and conceived the study, developed the theoretical analysis, implemented and validated the simulations, analyzed the data, prepared the figures and repository, and wrote the manuscript and Supplementary Information.

Declaration of AI-assisted tools. ChatGPT (OpenAI) was used for language polishing and for assistance with drafting, debugging, and checking Python scripts for numerical post-processing and figure preparation. The author independently reviewed and verified the code, calculations, figures, references, and manuscript text, and takes full responsibility for the analysis, results, and conclusions.

Acknowledgments. The author thanks Roderich Tumulka for discussions, critical questions, and invaluable input. The numerical calculations used computing resources provided by PC2, NHR@ZIB, and bwForCluster Helix. The author acknowledges support by the state of Baden-Württemberg through bwHPC Cluster and the German Research Foundation (DFG) through grant INST 35/1597-1 FUGG.

References

- [1] Muga, J. G. & Leavens, C. R. Arrival time in quantum mechanics. *Phys. Rep.* **338**, 353–438 (2000). doi:10.1016/S0370-1573(00)00047-8.
- [2] Allcock, G. R. The time of arrival in quantum mechanics. II. The individual measurement. *Ann. Phys. (N.Y.)* **53**, 286–310 (1969). doi:10.1016/0003-4916(69)90252-8.
- [3] Vona, N., Hinrichs, G. & Dürr, D. What does one measure when one measures the arrival time of a quantum particle? *Phys. Rev. Lett.* **111**, 220404 (2013). doi:10.1103/PhysRevLett.111.220404.
- [4] Das, S. & Dürr, D. Arrival time distributions of spin-1/2 particles. *Sci. Rep.* **9**, 2242 (2019). doi:10.1038/s41598-018-38261-4.
- [5] Tumulka, R. Distribution of the time at which an ideal detector clicks. *Ann. Phys. (N.Y.)* **442**, 168910 (2022). doi:10.1016/j.aop.2022.168910.
- [6] Tumulka, R. Absorbing boundary condition as limiting case of imaginary potentials. *Commun. Theor. Phys.* **75**, 015103 (2023). doi:10.1088/1572-9494/ac9bea.
- [7] Selstø, S. Absorption and analysis of unbound quantum particles one by one. *Phys. Rev. A* **103**, 012812 (2021). doi:10.1103/PhysRevA.103.012812.
- [8] Soffer, A., Stucchio, C. & Tran, M.-B. *Time Dependent Phase Space Filters: A Stable Absorbing Boundary Condition* (Springer, Singapore, 2023). doi:10.1007/978-981-19-6818-1.
- [9] Goldstein, S., Tumulka, R. & Zanghì, N. Arrival times versus detection times. *Found. Phys.* **54**, 63 (2024). doi:10.1007/s10701-024-00798-y.
- [10] Goldstein, S., Tumulka, R. & Zanghì, N. On the spin dependence of detection times and the nonmeasurability of arrival times. *Sci. Rep.* **14**, 3775 (2024). arXiv:2309.11835.
- [11] Cavendish, W. & Das, S. Absorbing detectors versus scattering theory. *Phys. Rev. A* **112**, 062217 (2025). doi:10.1103/332p-4nl1.
- [12] Halliwell, J. J. & Yearsley, J. M. Arrival times, complex potentials, and decoherent histories. *Phys. Rev. A* **79**, 062101 (2009). doi:10.1103/PhysRevA.79.062101.
- [13] Jozani, A. & Tumulka, R. Detection time distribution predicted using absorbing boundary conditions and imaginary potentials. arXiv:2603.22044 (2026).
- [14] Tumulka, R. Detection time distribution for Dirac particles. arXiv:1601.04571 (2016).
- [15] Frolov, L., Teufel, S. & Tumulka, R. Existence of Schrödinger evolution with absorbing boundary condition. *Math. Phys. Anal. Geom.* **28**, 27 (2025). doi:10.1007/s11040-025-09521-3.
- [16] Frolov, L. Detecting screens modeled by Schrödinger operators that generate C_0 contraction semigroups. *Rev. Math. Phys.*, to appear. arXiv:2506.00231 (2025).

- [17] See the Supplementary Information for numerical details, convergence tests, boundary-symbol and Duhamel derivations, reduced return bookkeeping, controls, parameter scans, and physical scale conversions.

Supplementary Information for “Spin–Momentum Impedance and Filtering by a Spin-Coupled Absorbing Boundary Condition”

Alireza Jozani¹

¹*Departments of Physics and Mathematics, Eberhard-Karls-Universität Tübingen,
Auf der Morgenstelle 10, 72076 Tübingen, Germany*

(Dated: June 24, 2026)

S1. SCOPE, NOTATION, AND RELATION TO THE MAIN TEXT

This Supplementary Information supports the main text’s central claim: the spin-coupled absorbing boundary condition (spinor ABC) acts as a local two-branch spin–momentum filter. The result is not a universal arrival-time law, but a detector-present boundary mechanism. For a tangential Fourier wave vector $\boldsymbol{\xi}$ at the detecting roof and detector impedance parameter κ , the spinor absorbing boundary has the two branches

$$\lambda_{\pm}(\boldsymbol{\xi}) = i\kappa \pm |\boldsymbol{\xi}|. \quad (\text{S1})$$

In a harmonic waveguide the transverse oscillator length is $\ell_{\perp} = \omega^{-1/2}$, so the typical tangential scale sampled at the roof is

$$|\boldsymbol{\xi}| \sim \ell_{\perp}^{-1} \sim \sqrt{\omega}. \quad (\text{S2})$$

Therefore transverse confinement enters the detector-present timing statistics through the local boundary symbol.

The organization is as follows. Section [S2](#) gives the numerical model, grid scaling, and convergence checks. [S3](#) gives the full confinement sweep supporting Fig. 1 of the main text. [S4](#) investigates the finite-window bookkeeping for the restricted mean detection time statistic shown in Fig. 1(c). [S5](#) shows that the leading first-pass bulk flux is independent of ω . Sections [S6](#) and [S7](#) derive and test the boundary-symbol mechanism, including the Duhamel estimate and finite-grid diagnostics. [S8](#) gives a reduced flux-balance and finite-guide memory closure explaining how the local boundary scale enters finite-window moments. Sections [S9–S11](#) collect controls, parameter scans, and physical scales.

The physical detector observable throughout this Supplement is the roof flux, equivalently the norm loss of the nonunitary spinor-ABC evolution. Bohmian trajectories, when shown as histograms, are used only as Monte Carlo samples of this same detector-present flux law. They are not a separate no-detector arrival-time proposal.

S2. NUMERICAL MODEL, GRID SCALING, AND CONVERGENCE

We work in units $\hbar = m = 1$ and solve the two-component time-dependent Schrödinger equation in the finite box

$$\Omega_\omega = [0, L_x(\omega)] \times [0, L_y(\omega)] \times [0, L], \quad (\text{S3})$$

with

$$i\partial_t \Psi = \left[-\frac{1}{2}\Delta + \frac{1}{2}\omega^2((x - x_c)^2 + (y - y_c)^2) \right] \Psi, \quad \Psi = \begin{pmatrix} \psi_\uparrow \\ \psi_\downarrow \end{pmatrix}. \quad (\text{S4})$$

The initial state is a normalized factorized packet,

$$\Psi_0(x, y, z) = \chi_\omega(x, y) \phi_0(z) \eta, \quad (\text{S5})$$

where the transverse factor is the harmonic ground-state profile, restricted and normalized in the finite transverse box,

$$\chi_\omega(x, y) = \mathcal{N}_\perp(\omega) \exp\left[-\frac{\omega}{2}((x - x_c)^2 + (y - y_c)^2)\right], \quad (\text{S6})$$

and the longitudinal factor is the right-moving Gaussian

$$\phi_0(z) = \mathcal{N}_z \exp\left[-\frac{(z - z_c)^2}{4\sigma_\parallel^2} + ik_z(z - z_c)\right], \quad 0 < z < L. \quad (\text{S7})$$

The constants $\mathcal{N}_\perp(\omega)$ and $\mathcal{N}_z$ normalize the factors on their finite numerical domains; equivalently, the full discrete state is normalized to unit norm before time evolution. Unless stated otherwise,

$$\eta = \begin{pmatrix} \cos(\theta/2) \\ e^{i\varphi} \sin(\theta/2) \end{pmatrix}, \quad \theta = 0, \quad \varphi = 0. \quad (\text{S8})$$

In the confinement sweep only ω is varied; the longitudinal packet, detector parameter, initial spinor, grid prescription, and observation window are fixed as listed in Table S1. The lower face $z = 0$ is a reflecting hard wall, implemented as a Dirichlet boundary in the finite-difference scheme. The artificial transverse walls are also Dirichlet. The detecting roof is

$$\Sigma_L = \{z = L\}, \quad (\text{S9})$$

where we impose the spinor absorbing boundary condition

$$(\boldsymbol{\sigma} \cdot \boldsymbol{\nabla})\Psi = i\kappa\sigma_z\Psi. \quad (\text{S10})$$

For the Pauli current this boundary condition gives the outward roof flux

$$j_z^P(x, y, L, t) = \kappa \Psi^\dagger \Psi(x, y, L, t) \geq 0. \quad (\text{S11})$$

Hence the detector-present detection-time density is

$$g(t; \omega) = \kappa \int_{\Sigma_L} \Psi^\dagger \Psi \, dx \, dy = -\frac{d}{dt} \|\Psi_t\|^2. \quad (\text{S12})$$

We denote the survival probability by

$$S(t; \omega) = \|\Psi_t\|^2. \quad (\text{S13})$$

For a finite observation window T_{obs} we use

$$D_{T_{\text{obs}}}(\omega) = \int_0^{T_{\text{obs}}} g(t; \omega) \, dt \quad (\text{S14})$$

and

$$\mu^*(T_{\text{obs}}; \omega) = \int_0^{T_{\text{obs}}} S(t; \omega) \, dt = \mathbb{E}[\min(\tau, T_{\text{obs}})], \quad (\text{S15})$$

where τ is the random detector click time. The restricted mean detection time μ^* is used because for strong confinement a substantial fraction of the probability can remain undetected at the end of the finite simulation window.

The numerical evolution uses a Cartesian finite-difference Crank–Nicolson scheme with GMRES solves for the resulting non-Hermitian sparse systems. The spinor ABC is imposed by ghost-point elimination at the roof, producing the top-layer spinor coupling associated with the tangential derivatives in the boundary condition; this follows the implementation described in Ref. [13] of the main text. Here we report only the present parameter choices, grid scaling, convergence checks, and diagnostics.

The transverse box is scaled with the oscillator length

$$\ell_\perp = \omega^{-1/2}. \quad (\text{S16})$$

Specifically,

$$L_x(\omega) = L_y(\omega) = \ell_{\text{box}} \ell_\perp = \frac{\ell_{\text{box}}}{\sqrt{\omega}}, \quad \ell_{\text{box}} = 10. \quad (\text{S17})$$

With $N_x = N_y = N_\perp$, the transverse mesh width satisfies

$$\frac{h_\perp}{\ell_\perp} = \frac{\ell_{\text{box}}}{N_\perp}. \quad (\text{S18})$$

Therefore both the wall distance in oscillator units and the transverse resolution in oscillator units are fixed across the confinement sweep. This prevents the observed ω -dependence from being a lateral-wall or transverse-resolution artifact.

TABLE S1. Reference parameters for the confinement sweep shown in the main text and in Fig. S1. Only ω is varied.

Quantity	Value
Transverse box size	$L_x = L_y = \ell_{\text{box}}/\sqrt{\omega}$, $\ell_{\text{box}} = 10$
Longitudinal box size	$L = L_z = 20$
Transverse center	$x_c = L_x/2$, $y_c = L_y/2$
Initial longitudinal center	$z_c = 10$
Detector parameter	$\kappa = \pi$
Longitudinal wave number	$k_z = \pi$
Longitudinal width	$\sigma_{\parallel} = 0.5$
Initial spinor	$\theta = 0$, $\phi = 0$, unless otherwise stated
Reference transverse grid	$N_x = N_y = 100$
Reference longitudinal grid	$N_z = 1500$
Reference time step	$\Delta t = 2 \times 10^{-4}$
Observation window	$T_{\text{obs}} = 20$
Varied parameter	ω

As a representative convergence check, we use the strongest confinement shown in the main text, $\omega = 300$. This is the most demanding point of the main sweep because the transverse oscillator length is smallest and the delayed oscillatory sector is largest. We independently refine the transverse grid, the longitudinal grid, and the time step, while monitoring D_{20} and $\mu^*(20)$.

TABLE S2. Representative convergence checks for $\mu^*(20)$ at $\omega = 300$. The first row is the reference run used in Fig. S1. The relative change is $\epsilon_{\text{rel}} = |\mu_{\text{ref}}^* - \mu_{\text{refined}}^*|/|\mu_{\text{refined}}^*| \times 100\%$. All runs use $\kappa = k_z = \pi$, $\sigma_{\parallel} = 0.5$, and $T_{\text{obs}} = 20$.

Run	$N_x = N_y$	N_z	Δt	D_{20}	$\mu^*(20)$
Reference	100	1500	2×10^{-4}	0.42	14.783
Transverse refinement	120	1500	2×10^{-4}	0.42	14.791
Longitudinal refinement	100	2000	2×10^{-4}	0.42	14.841
Time-step refinement	100	1500	1×10^{-4}	0.42	14.782

The largest change in $\mu^*(20)$ in Table S2 occurs under longitudinal refinement and is below 0.4%. This is small compared with the confinement-induced change over the sweep and does not affect the qualitative conclusion: increasing ω suppresses the prompt peak, reduces the finite-window detected fraction, and enhances the delayed detection time sector.

S3. FULL CONFINEMENT SWEEP SUPPORTING FIG. 1

Here the full confinement sweep in Fig. 1 of the main text is given. Only ω is varied. The longitudinal packet, detector parameter, initial spinor, and observation window are fixed as in Table S1. The plotted density is the detector-present roof-flux density $g(t; \omega)$. The red curve is the detector-free one-dimensional Gaussian comparison current through $z = L$ for the same longitudinal packet, with the transverse factor integrated out. It is not a detector model and is not used in the fit; it is the analogue of the comparison curve used in Ref. [13].

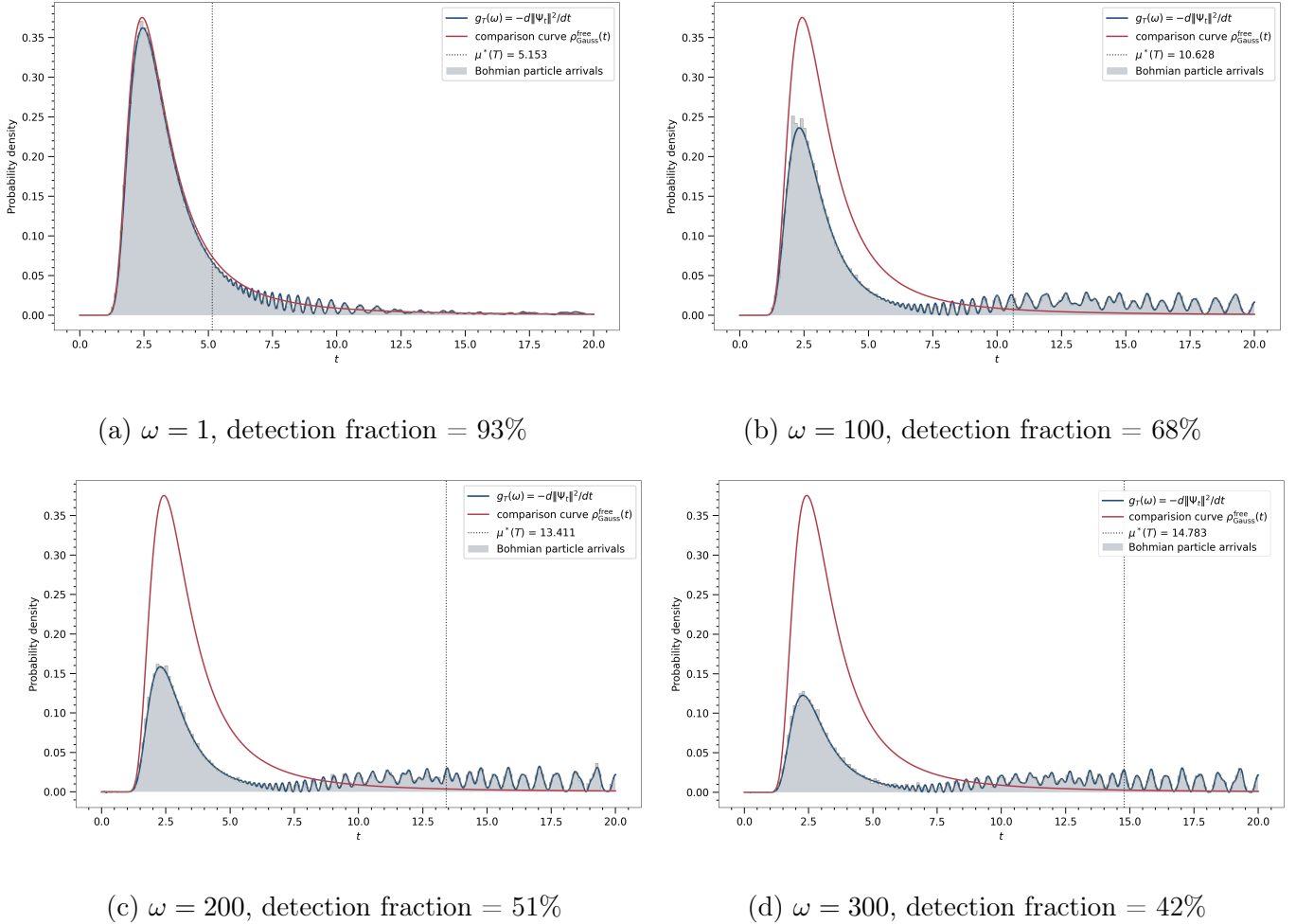


FIG. S1. Full confinement sweep for the spinor absorbing boundary. Only ω is varied. Increasing ω narrows the harmonic transverse state, suppresses the prompt roof-flux peak, reduces the detected fraction, and shifts weight into a delayed oscillatory sector. Blue: detector-present roof-flux density $g(t; \omega)$; red: detector-free Gaussian comparison current; gray: Bohmian sampling histograms of the same detector-present flux law.

The finite-window mean detection time for several ω is fitted by

$$\mu^*(20; \omega) \simeq 4.084 + 0.638\sqrt{\omega}. \quad (\text{S19})$$

This fit is empirical and finite-window dependent. The coefficients are not universal; the robust feature is the local boundary scale $R = |\xi| \sim \sqrt{\omega}$, derived in Sec. S6.

S4. FINITE-WINDOW BOOKKEEPING FOR THE STATISTIC IN FIG. 1(C)

This section gives the finite-window bookkeeping used in the main text and a diagnostic split of the computed roof-flux signal into a prompt peak and a delayed sector. The split is only a numerical diagnostic of the detector-present roof flux $g(t; \omega)$. It is not an exact microscopic decomposition of the TDSE solution into first-pass and return amplitudes.

S4.1. Conditional finite-window density and RMDT identity

Let T_{obs} denote the numerical observation horizon and let τ be the random detection time. For compactness, write $T = T_{\text{obs}}$ in this section. The detected fraction by time T is

$$D_T(\omega) = \mathbb{P}(\tau \leq T) = \int_0^T g(t; \omega) dt. \quad (\text{S20})$$

When $D_T(\omega) > 0$, the detection-time density conditioned on detection inside the numerical window is

$$f_T(t; \omega) = \frac{g(t; \omega)}{D_T(\omega)}, \quad 0 \leq t \leq T. \quad (\text{S21})$$

Its conditional mean is

$$\bar{\tau}_T^{\text{cond}}(\omega) = \int_0^T t f_T(t; \omega) dt. \quad (\text{S22})$$

The finite-window statistic used in the main text is the restricted mean detection time

$$\mu^*(T; \omega) = \int_0^T S(t; \omega) dt, \quad S(t; \omega) = \|\Psi_t\|^2. \quad (\text{S23})$$

Using $g(t; \omega) = -dS(t; \omega)/dt$, integration by parts gives

$$\begin{aligned} \mu^*(T; \omega) &= TS(T; \omega) + \int_0^T t g(t; \omega) dt \\ &= [1 - D_T(\omega)]T + D_T(\omega) \bar{\tau}_T^{\text{cond}}(\omega). \end{aligned} \quad (\text{S24})$$

Therefore μ^* contains two finite-window effects: the conditional timing of the detected part and the probability still undetected at the final time. Equivalently,

$$\bar{\tau}_T^{\text{cond}}(\omega) = \frac{\mu^*(T; \omega) - [1 - D_T(\omega)]T}{D_T(\omega)}. \quad (\text{S25})$$

This identity is used below to keep the conditional early/late diagnostic separate from the restricted mean detection time (RMDT) fit shown in the main text.

S4.2. Common diagnostic cutoff

The early/late split uses one cutoff time t_{cut} , common to all values of ω . The cutoff is not fitted from the roof-flux data and is not chosen separately for each run. It is used only to define a fixed diagnostic window before the earliest estimated contamination by a full bottom-return cycle. Let

$$\Sigma_i = \{z = z_i\} \quad (\text{S26})$$

be an interior monitoring plane below the detector. From the main upward-flux peak times on two such planes, we estimate an effective axial speed $v_{\text{est}}(\omega)$. If $t_{\text{pk,up}}^{(i)}(\omega)$ is the main upward peak time on Σ_i , then the estimated time of the first roof-return signal at that plane is

$$t_{\text{return}}^{(i)}(\omega) = t_{\text{pk,up}}^{(i)}(\omega) + \frac{2d_R^{(i)}}{v_{\text{est}}(\omega)}, \quad d_R^{(i)} = z_{\text{roof}} - z_i. \quad (\text{S27})$$

A subsequent signal from a full bottom-return cycle would require an additional round trip from the monitoring plane to the bottom and back. We therefore define no-return time

$$t_{\text{nr}}^{(i)}(\omega) = t_{\text{return}}^{(i)}(\omega) + \gamma \frac{2d_B^{(i)}}{v_{\text{est}}(\omega)}, \quad d_B^{(i)} = z_i - z_{\text{bottom}}, \quad \gamma = 0.95. \quad (\text{S28})$$

The parameter γ is not a reflection coefficient and is not used to model the dynamics. It only gives a conservative timing estimate for the diagnostic split. For the closer monitoring plane Σ_1 , the estimated no-return times are

ω	1	50	100	150	200	300
$t_{\text{nr}}^{(1)}(\omega)$	13.58	7.55	6.02	5.69	4.46	3.69.

(S29)

The smallest value is about 3.69. We therefore use the common cutoff

$$t_{\text{cut}} = 3.6. \quad (\text{S30})$$

S4.3. Early/late identity and numerical values

Define the early and late intervals

$$I_E = \{0 \leq t \leq t_{\text{cut}}\}, \quad I_L = \{t_{\text{cut}} < t \leq T\}. \quad (\text{S31})$$

The late conditional weight is

$$w_L(\omega) = \int_{t_{\text{cut}}}^T f_T(t; \omega) dt, \quad w_E(\omega) = 1 - w_L(\omega). \quad (\text{S32})$$

The early and late conditional means are

$$t_E(\omega) = \frac{1}{w_E(\omega)} \int_0^{t_{\text{cut}}} t f_T(t; \omega) dt, \quad (\text{S33})$$

and

$$t_L(\omega) = \frac{1}{w_L(\omega)} \int_{t_{\text{cut}}}^T t f_T(t; \omega) dt. \quad (\text{S34})$$

Then the conditional finite-window mean detection satisfies the exact identity

$$\begin{aligned} \bar{\tau}_T^{\text{cond}}(\omega) &= w_E(\omega)t_E(\omega) + w_L(\omega)t_L(\omega) \\ &= t_E(\omega) + w_L(\omega)[t_L(\omega) - t_E(\omega)]. \end{aligned} \quad (\text{S35})$$

We then define the delayed-sector contribution to the conditional mean by

$$\Delta t_L^{\text{cond}}(\omega) := \bar{\tau}_T^{\text{cond}}(\omega) - t_E(\omega) = w_L(\omega)[t_L(\omega) - t_E(\omega)]. \quad (\text{S36})$$

TABLE S3. Early/late split of the finite-window detector statistics for $T = 20$ and $t_{\text{cut}} = 3.6$. Here D_T is the detected fraction, $\bar{\tau}_T^{\text{cond}}$ is the mean conditioned on detection before T , and $w_L, t_L, \Delta t_L^{\text{cond}}$ are computed with respect to the conditional density f_T .

ω	D_T	$\bar{\tau}_T^{\text{cond}}$	$\mu^*(T)$	t_E	w_L	t_L	Δt_L^{cond}
1	0.925	3.95	5.15	2.59	0.37	6.18	1.35
50	0.788	5.00	8.18	2.54	0.39	8.77	2.46
100	0.670	6.00	10.62	2.51	0.45	10.18	3.48
150	0.577	6.63	12.29	2.51	0.49	10.82	4.12
200	0.507	7.01	13.41	2.50	0.52	11.17	4.51
300	0.416	7.45	14.78	2.49	0.54	11.55	4.95

Fitting the displayed diagnostic quantities as affine functions of $\sqrt{\omega}$ gives

$$t_E(\omega) \approx 2.52 \pm 0.04, \quad (\text{S37})$$

$$\Delta t_L^{\text{cond}}(\omega) \approx (1.07 \pm 0.18) + (0.234 \pm 0.015)\sqrt{\omega}, \quad (\text{S38})$$

$$\bar{\tau}_T^{\text{cond}}(\omega) \approx (3.66 \pm 0.18) + (0.228 \pm 0.015)\sqrt{\omega}, \quad (\text{S39})$$

$$\mu^*(T; \omega) \approx (4.38 \pm 0.36) + (0.618 \pm 0.031)\sqrt{\omega}. \quad (\text{S40})$$

The fit in Eq. (S40) uses the rounded diagnostic values displayed in Table S3, whereas Eq. (S19) above and Fig. 1(c) of the main text use the unrounded full confinement-sweep data.

The near constancy of t_E should not be read as the absence of a first-encounter boundary effect. The spinor ABC already reads the transverse scale during the first near-roof interaction, through the local boundary scale $R = |\xi| \sim \sqrt{\omega}$. What is nearly constant is only the conditional mean time of those detections that remain in the early window.

The stronger prompt-sector signature is instead the loss of early detected weight. Defining

$$D_E(\omega) := D_T(\omega) [1 - w_L(\omega)], \quad (\text{S41})$$

the data give $D_E(1) \simeq 0.58$ and $D_E(300) \simeq 0.19$. Thus increasing ω strongly suppresses the prompt roof-flux contribution, even though the mean time t_E of the remaining prompt detections changes only weakly. The delayed sector should therefore be understood as the finite-box continuation and magnification, in time moments, of the same spinor-ABC boundary filtering already imposed at the first detector encounter.

Thus, the early conditional mean changes only weakly over the scan and does not carry the dominant square-root trend. The delayed conditional contribution grows approximately affinely in $\sqrt{\omega}$ over this finite window. The RMDT trend, however, must not be identified directly with Δt_L^{cond} . Combining Eqs. (S24) and (S36) gives

$$\mu^*(T; \omega) = D_T(\omega) [t_E(\omega) + \Delta t_L^{\text{cond}}(\omega)] + [1 - D_T(\omega)]T. \quad (\text{S42})$$

Therefore, the fitted coefficient of the RMDT receives contributions from the weak residual variation of $t_E(\omega)$, the growth of $\Delta t_L^{\text{cond}}(\omega)$, and the decreasing detected fraction $D_T(\omega)$ through the survival term $[1 - D_T(\omega)]T$. The late sector is not a separate physical mechanism. It is the finite-box delayed continuation of the same spinor-ABC boundary-filtering mechanism derived in Sec. S6 and represented effectively by the reduced flux-balance closure in Sec. S8.

S5. FIRST-PASS BULK INDEPENDENCE OF ω

We begin with an analytic bulk check. Before the packet has significantly touched the absorbing roof, ordinary first-pass propagation cannot generate the observed confinement dependence on ω . In a separable, spin-independent bulk, changing ω only changes the normalized transverse factor; after cross-section integration, the leading Pauli flux is purely longitudinal and independent of ω .

We now show this explicitly. Let $z_b = L - \delta$ be an interior counting plane below the detector, and let

$$\Sigma_b = \{(x, y, z) : z = z_b\}. \quad (\text{S43})$$

In the region below the roof, assume the first-pass separable bulk Hamiltonian

$$H_{\text{bulk}} = H_{xy}(\omega) + H_z, \quad (\text{S44})$$

with

$$H_{xy}(\omega) = -\frac{1}{2}(\partial_x^2 + \partial_y^2) + V_{xy}(x, y; \omega), \quad H_z = -\frac{1}{2}\partial_z^2. \quad (\text{S45})$$

We also assume that the absorbing roof has not yet significantly modified the state at the interior plane. For a prepared state with spatially constant spinor η , the auxiliary first-pass bulk field is

$$\Psi_{\text{fp}}(x, y, z, t) = \chi_\omega(x, y, t) \phi_{\text{fp}}(z, t) \eta, \quad \rho_{\text{fp}} = |\Psi_{\text{fp}}|^2 = |\chi_\omega(x, y, t)|^2 |\phi_{\text{fp}}(z, t)|^2. \quad (\text{S46})$$

Here “first-pass” means the separable, spin-independent bulk evolution before appreciable feedback from the absorbing roof. The longitudinal factor is initialized by $\phi_{\text{fp}}(z, 0) = \phi_0(z)$, and Ψ_{fp} is not the full spinor-ABC solution. It is used only to test whether ordinary bulk propagation can generate the observed ω -dependence. The spin density of this auxiliary field is

$$\mathbf{S}_{\text{fp}} = \Psi_{\text{fp}}^\dagger \boldsymbol{\sigma} \Psi_{\text{fp}} = \rho_{\text{fp}} \mathbf{s}_\eta, \quad \mathbf{s}_\eta = \eta^\dagger \boldsymbol{\sigma} \eta. \quad (\text{S47})$$

The corresponding Pauli current is

$$\mathbf{j}_{\text{fp}}^P = \mathbf{j}_{\text{fp}}^{\text{conv}} + \frac{1}{2} \nabla \times \mathbf{S}_{\text{fp}}, \quad \mathbf{j}_{\text{fp}}^{\text{conv}} = \text{Im}(\Psi_{\text{fp}}^\dagger \nabla \Psi_{\text{fp}}). \quad (\text{S48})$$

Since $\mathbf{s}_\eta$ is constant in the first-pass separable approximation,

$$(\nabla \times \mathbf{S}_{\text{fp}})_z = \partial_x(\rho_{\text{fp}} s_{\eta,y}) - \partial_y(\rho_{\text{fp}} s_{\eta,x}). \quad (\text{S49})$$

Therefore

$$\int_{\Sigma_b} (\nabla \times \mathbf{S}_{\text{fp}})_z dx dy = \int_{\partial \Sigma_b} (\mathbf{S}_{\text{fp},y} \nu_x - \mathbf{S}_{\text{fp},x} \nu_y) d\ell = 0, \quad (\text{S50})$$

where $\boldsymbol{\nu} = (\nu_x, \nu_y)$ is the outward unit normal to $\partial \Sigma_b$ in the transverse plane. The last equality follows from the lateral boundary conditions in the finite box, or from transverse decay in the ideal waveguide. Hence the plane-integrated Pauli flux equals the plane-integrated convective flux:

$$\int_{\Sigma_b} j_{\text{fp},z}^P dx dy = \int_{\Sigma_b} j_{\text{fp},z}^{\text{conv}} dx dy. \quad (\text{S51})$$

Define the first-pass bulk flux through Σ_b by

$$\Phi_{\text{bulk}}(t) = \int_{\Sigma_b} j_{\text{fp},z}^P(\mathbf{r}, t) dA. \quad (\text{S52})$$

Using Eq. (S46),

$$\begin{aligned} \Phi_{\text{bulk}}(t) &= \int_{\Sigma_b} \text{Im}(\Psi_{\text{fp}}^\dagger \partial_z \Psi_{\text{fp}}) dA \\ &= \left(\int |\chi_\omega(x, y, t)|^2 dx dy \right) \text{Im}(\phi_{\text{fp}}^* \partial_z \phi_{\text{fp}})(z_b, t). \end{aligned} \quad (\text{S53})$$

Since $H_{xy}(\omega)$ is self-adjoint with the chosen transverse boundary conditions, the transverse factor remains normalized. Therefore

$$\Phi_{\text{bulk}}(t) = \text{Im}(\phi_{\text{fp}}^* \partial_z \phi_{\text{fp}})(z_b, t), \quad (\text{S54})$$

which is independent of ω and independent of the detector parameter κ . Accordingly, the first-pass flux-averaged crossing time

$$\bar{\tau}_{\text{bulk}} = \frac{\int_0^\infty t \Phi_{\text{bulk}}(t) dt}{\int_0^\infty \Phi_{\text{bulk}}(t) dt} \quad (\text{S55})$$

is ω -independent under the stated assumptions. This does not claim that the full finite-box detector signal is ω -independent. It only shows that the leading first-pass bulk transport does not generate the confinement trend; the effect must enter through the spinor absorbing boundary.

S6. BOUNDARY-SYMBOL MECHANISM

This section studies the detailed boundary-symbol calculation used in the main text. The calculation is a local probe of the spinor ABC, not a second absorber and not a physical layer below the detector. The physical detector observable remains the roof flux at $z = L$. At the roof the spinor ABC is

$$(\boldsymbol{\sigma} \cdot \boldsymbol{\nabla})\Psi = i\kappa\boldsymbol{\sigma}_z\Psi, \quad \Psi = \begin{pmatrix} \psi_\uparrow \\ \psi_\downarrow \end{pmatrix}. \quad (\text{S56})$$

In components,

$$\begin{aligned} \partial_z\psi_\uparrow &= i\kappa\psi_\uparrow - (\partial_x - i\partial_y)\psi_\downarrow, \\ \partial_z\psi_\downarrow &= i\kappa\psi_\downarrow + (\partial_x + i\partial_y)\psi_\uparrow. \end{aligned} \quad (\text{S57})$$

Equivalently, the roof condition is the first-order normal boundary relation

$$\partial_z\Psi|_{z=L} = \mathcal{C}\Psi|_{z=L}, \quad \mathcal{C} = \begin{pmatrix} i\kappa & -D_- \\ D_+ & i\kappa \end{pmatrix}, \quad D_\pm = \partial_x \pm i\partial_y. \quad (\text{S58})$$

Therefore the detector does not impose only a scalar normal impedance. It couples the normal boundary relation to tangential derivatives of the roof trace.

S6.1. Exact near-roof identity

We now isolate the local normal response generated by the boundary relation itself. Fix a time t and take the roof trace $\Psi(\cdot, L, t)$ as given. We use the homogeneous relation $\partial_z\Psi = \mathcal{C}\Psi$ only to continue this trace a small auxiliary distance inward, to $z = L - \epsilon$ with $\epsilon > 0$. This continuation is a local probe of the boundary symbol, not a physical detector layer; the true TDSE solution need not satisfy $\partial_z\Psi = \mathcal{C}\Psi$ away from the roof. Define the boundary-condition defect/error in a small interval $L - \delta < z \leq L$ by

$$\mathcal{E}_{\text{bc}}(z, t) := \partial_z\Psi(\cdot, z, t) - \mathcal{C}\Psi(\cdot, z, t). \quad (\text{S59})$$

The spinor ABC implies

$$\mathcal{E}_{\text{bc}}(L, t) = 0. \quad (\text{S60})$$

After tangential Fourier transform, each mode satisfies

$$\partial_z\widehat{\Psi}(\boldsymbol{\xi}, z, t) = \mathcal{C}(\boldsymbol{\xi})\widehat{\Psi}(\boldsymbol{\xi}, z, t) + \widehat{\mathcal{E}}_{\text{bc}}(\boldsymbol{\xi}, z, t). \quad (\text{S61})$$

Variation of constants, integrated backward from L to z , gives the exact near-roof identity

$$\widehat{\Psi}(\boldsymbol{\xi}, z, t) = e^{(z-L)\mathcal{C}(\boldsymbol{\xi})}\widehat{\Psi}(\boldsymbol{\xi}, L, t) - \int_z^L e^{(z-s)\mathcal{C}(\boldsymbol{\xi})}\widehat{\mathcal{E}}_{\text{bc}}(\boldsymbol{\xi}, s, t) ds. \quad (\text{S62})$$

The first term is the homogeneous normal continuation generated by the spinor ABC. The second term is the Duhamel remainder. We denote the roof trace by

$$W_\omega(\boldsymbol{\xi}, t) := \widehat{\Psi}_\omega(\boldsymbol{\xi}, L, t). \quad (\text{S63})$$

For $z = L - \epsilon$, the Duhamel remainder is

$$\widehat{\mathcal{E}}_{\text{Duh}}(\boldsymbol{\xi}, L - \epsilon, t) := - \int_0^\epsilon e^{-(\epsilon-u)\mathcal{C}(\boldsymbol{\xi})} \widehat{\mathcal{E}}_{\text{bc}}(\boldsymbol{\xi}, L - u, t) du. \quad (\text{S64})$$

This is the same remainder term denoted by $\widehat{\mathcal{R}}_{\text{Duh}}$ in the main text. Under the regularity assumption quantified in Sec. S7.1, this remainder is $O(\epsilon^2)$ modewise as $\epsilon \rightarrow 0$. Hence it has no zeroth- or first-order contribution at the roof. The leading inward normal response is therefore determined by the homogeneous boundary symbol.

S6.2. Tangential symbol and spin-momentum branches

The tangential Fourier transform is used only to expose the local symbol of the boundary operator. In the finite-difference calculation, the analogous object is the discrete tangential symbol used in Sec. S7. For a tangential Fourier mode

$$\boldsymbol{\xi} = (\xi_x, \xi_y), \quad \xi_\pm = \xi_x \pm i\xi_y, \quad (\text{S65})$$

Eq. (S58) gives

$$\mathcal{C}(\boldsymbol{\xi}) = \begin{pmatrix} i\kappa & -i\xi_- \\ i\xi_+ & i\kappa \end{pmatrix}. \quad (\text{S66})$$

Separate the scalar impedance from the tangential spin coupling:

$$\mathcal{J}(\boldsymbol{\xi}) := \mathcal{C}(\boldsymbol{\xi}) - i\kappa I. \quad (\text{S67})$$

Then

$$\mathcal{J}(\boldsymbol{\xi}) = (\hat{\mathbf{z}} \times \boldsymbol{\xi}) \cdot \boldsymbol{\sigma} = -\xi_y \sigma_x + \xi_x \sigma_y, \quad \mathcal{J}(\boldsymbol{\xi})^2 = R^2 I, \quad R = |\boldsymbol{\xi}|. \quad (\text{S68})$$

For $R > 0$, define

$$\Gamma(\boldsymbol{\xi}) = \frac{\mathcal{J}(\boldsymbol{\xi})}{R} = (\hat{\mathbf{z}} \times \hat{\boldsymbol{\xi}}) \cdot \boldsymbol{\sigma}, \quad \Gamma(\boldsymbol{\xi})^2 = I. \quad (\text{S69})$$

The zero mode $R = 0$ is degenerate and carries no $R\Gamma$ contribution; it is understood by continuity. Therefore

$$\mathcal{C}(\boldsymbol{\xi}) = i\kappa I + R\Gamma(\boldsymbol{\xi}). \quad (\text{S70})$$

The two boundary eigenbranches are

$$\lambda_\pm(\boldsymbol{\xi}) = i\kappa \pm R, \quad (\text{S71})$$

with projectors

$$\Pi_{\pm}(\boldsymbol{\xi}) = \frac{1}{2}(I \pm \Gamma(\boldsymbol{\xi})). \quad (\text{S72})$$

The projectors select two tangential spin–momentum branches, with in-plane spin direction perpendicular to the tangential momentum. This is the key local difference between the spinor ABC and a scalar absorbing boundary.

S6.3. Auxiliary normal response

Since $\Gamma^2 = I$, the homogeneous exponential in Eq. (S62) is explicit:

$$\begin{aligned} e^{-\epsilon\mathcal{C}(\boldsymbol{\xi})} &= e^{-i\kappa\epsilon} e^{-\epsilon R\Gamma(\boldsymbol{\xi})} \\ &= e^{-i\kappa\epsilon} [e^{-R\epsilon}\Pi_+(\boldsymbol{\xi}) + e^{R\epsilon}\Pi_-(\boldsymbol{\xi})]. \end{aligned} \quad (\text{S73})$$

Putting $z = L - \epsilon$ in Eq. (S62) gives

$$\widehat{\Psi}_{\omega}(\boldsymbol{\xi}, L - \epsilon, t) = e^{-i\kappa\epsilon} B_{\text{br}}(\epsilon, \boldsymbol{\xi}) W_{\omega}(\boldsymbol{\xi}, t) + \widehat{\mathcal{R}}_{\text{Duh}}(\boldsymbol{\xi}, L - \epsilon, t), \quad (\text{S74})$$

where the two-branch normal-response map is

$$B_{\text{br}}(\epsilon, \boldsymbol{\xi}) = e^{-R\epsilon}\Pi_+(\boldsymbol{\xi}) + e^{R\epsilon}\Pi_-(\boldsymbol{\xi}). \quad (\text{S75})$$

This is the matrix-valued branch filter acting on spinor amplitudes. The factor $e^{-R\epsilon}\Pi_+$ is the boundary-induced evanescent normal factor, with length scale R^{-1} , and the term $e^{R\epsilon}\Pi_-$ is the companion branch of the same auxiliary inward continuation. It is not physical amplification, not an absorbed component, and not a second detector model. Using Eq. (S72),

$$B_{\text{br}}(\epsilon, \boldsymbol{\xi}) = \cosh(R\epsilon)I - \sinh(R\epsilon)\Gamma(\boldsymbol{\xi}) = e^{-R\epsilon\Gamma(\boldsymbol{\xi})}. \quad (\text{S76})$$

Thus the branch structure is not visible as a multiplicative suppression of the roof density itself, since $B_{\text{br}}(0, \boldsymbol{\xi}) = I$. It enters through the first inward normal response:

$$\partial_{\epsilon} B_{\text{br}}(\epsilon, \boldsymbol{\xi})|_{\epsilon=0} = -R\Gamma(\boldsymbol{\xi}) = -\mathcal{J}(\boldsymbol{\xi}). \quad (\text{S77})$$

This is the local sense in which the spinor ABC filters tangential spin–momentum: the same tangential radius $R = |\boldsymbol{\xi}|$ that labels the roof mode sets the scale of the first normal response at the boundary.

For the harmonic transverse ground state, $\ell_{\perp} = \omega^{-1/2}$, so the typical tangential wave number sampled by the roof trace satisfies¹

$$R = |\boldsymbol{\xi}| \sim \ell_{\perp}^{-1} \sim \sqrt{\omega}. \quad (\text{S78})$$

¹ Equivalently, for normalized transverse oscillator modes centered at the trap center, $D_{\pm}\chi_{00}^{(\omega)} = -\sqrt{\omega}\chi_{\pm}^{(\omega)}$, $\chi_{\pm}^{(\omega)} = \frac{\chi_{10}^{(\omega)} \pm i\chi_{01}^{(\omega)}}{\sqrt{2}}$, up to phase conventions. This is the oscillator-basis version of the same boundary-symbol scale $R = |\boldsymbol{\xi}| \sim \sqrt{\omega}$.

A complementary local stationary viewpoint gives the same branch structure. If one freezes a tangential branch satisfying $\Pi_{\pm}\Psi = \Psi$, the roof condition reduces to

$$\partial_z \psi_{\pm} = (i\kappa \pm R)\psi_{\pm}. \quad (\text{S79})$$

Therefore even the local no-click matching problem is governed by the non-scalar pair $i\kappa \pm R$, not by a single scalar Robin impedance. In a frozen half-space plane-wave check this corresponds to a matrix no-click reflection amplitude of the form

$$\text{Ref}_{\text{loc}}(k_n, \boldsymbol{\xi}) = r_+(k_n, R)\Pi_+(\boldsymbol{\xi}) + r_-(k_n, R)\Pi_-(\boldsymbol{\xi}), \quad (\text{S80})$$

where $k_n > 0$ is the normal wave number. In that frozen local model the two branch intensities are equal, but the branch phases and projectors remain different. We do not use this stationary viewpoint as a fit model or as a scattering theory of the finite guide. The full phase-resolved scattering problem, including oscillator-channel evanescent components, is separate from the finite-window roof-flux mechanism studied here.²

S6.4. Covariance form of the local boundary source

The previous subsection identifies the local boundary scale. We now express its first-order effect on a normalized auxiliary time moment in covariance form. This auxiliary moment is only a diagnostic of the local boundary response; the physical detector statistic remains the roof flux at $z = L$.

Since the scalar phase $e^{-i\kappa\epsilon}$ drops out of densities, the relevant first-order density response is obtained after removing this phase. By Eq. (S74) and the Duhamel estimate of Sec. S7.1,

$$e^{i\kappa\epsilon}\widehat{\Psi}_{\omega}(\boldsymbol{\xi}, L - \epsilon, t) = B_{\text{br}}(\epsilon, \boldsymbol{\xi})W_{\omega}(\boldsymbol{\xi}, t) + O_{\text{Duh}}(\epsilon^2). \quad (\text{S81})$$

Thus the exact phase-removed trace and the homogeneous branch probe have the same zeroth and first inward variations at the roof. We therefore define

$$\widetilde{W}_{\omega, \epsilon}(\boldsymbol{\xi}, t) := B_{\text{br}}(\epsilon, \boldsymbol{\xi})W_{\omega}(\boldsymbol{\xi}, t), \quad (\text{S82})$$

and compute the first normal variation from this homogeneous branch probe. We keep the branch exponential exact, rather than replacing it by a finite-depth Taylor approximation. Since $B_{\text{br}}(\epsilon, \boldsymbol{\xi}) = e^{-\epsilon R\Gamma(\boldsymbol{\xi})}$ and $\Gamma(\boldsymbol{\xi})$ is Hermitian with $\Gamma^2 = I$, we have

$$B_{\text{br}}(\epsilon, \boldsymbol{\xi})^{\dagger}B_{\text{br}}(\epsilon, \boldsymbol{\xi}) = e^{-2\epsilon R\Gamma(\boldsymbol{\xi})} = \cosh(2R\epsilon)I - \sinh(2R\epsilon)\Gamma(\boldsymbol{\xi}). \quad (\text{S83})$$

² The author thanks Roderich Tumulka for discussions on stationary scattering viewpoints and evanescent boundary responses.

Therefore the auxiliary homogeneous boundary-layer spectral density is exactly

$$\rho_{\omega,\epsilon}^{\text{bl}}(\boldsymbol{\xi}, t) := \left| \widetilde{W}_{\omega,\epsilon}(\boldsymbol{\xi}, t) \right|^2 = W_{\omega}^{\dagger} e^{-2\epsilon R \Gamma(\boldsymbol{\xi})} W_{\omega}. \quad (\text{S84})$$

On the null set where $W_{\omega}^{\dagger} W_{\omega} = 0$, a_{ω} may be defined arbitrarily, e.g. $a_{\omega} = 0$. Writing

$$\rho_{\omega,0}^{\text{bl}}(\boldsymbol{\xi}, t) = W_{\omega}^{\dagger} W_{\omega}, \quad \text{and} \quad a_{\omega}(\boldsymbol{\xi}, t) := -2 \frac{W_{\omega}^{\dagger} \Gamma(\boldsymbol{\xi}) W_{\omega}}{W_{\omega}^{\dagger} W_{\omega}}, \quad (\text{S85})$$

we obtain the exact reweighting form

$$\rho_{\omega,\epsilon}^{\text{bl}}(\boldsymbol{\xi}, t) = \rho_{\omega,0}^{\text{bl}}(\boldsymbol{\xi}, t) b_{\omega,\epsilon}(\boldsymbol{\xi}, t), \quad (\text{S86})$$

where

$$b_{\omega,\epsilon}(\boldsymbol{\xi}, t) = \cosh(2R\epsilon) + \frac{a_{\omega}(\boldsymbol{\xi}, t)}{2} \sinh(2R\epsilon). \quad (\text{S87})$$

The quantity a_{ω} is real because Γ is Hermitian. Since the eigenvalues of Γ are ± 1 ,

$$|a_{\omega}(\boldsymbol{\xi}, t)| \leq 2. \quad (\text{S88})$$

In particular,

$$b_{\omega,0} = 1, \quad \partial_{\epsilon} b_{\omega,\epsilon}|_{\epsilon=0} = R a_{\omega}(\boldsymbol{\xi}, t). \quad (\text{S89})$$

No finite-depth smallness assumption $R\epsilon \ll 1$ is used here: the branch map has been kept exact, and only its derivative at the physical roof is taken. We now pass from this pointwise density reweighting to a normalized auxiliary time moment. Let

$$d\mu_{\omega}^{(0)}(\boldsymbol{\xi}, t) := \rho_{\omega,0}^{\text{bl}}(\boldsymbol{\xi}, t) d^2 \boldsymbol{\xi} dt, \quad Z_{\omega}^{(0)} := \int d\mu_{\omega}^{(0)}, \quad (\text{S90})$$

where the time integral is over the chosen finite observation window. Define the normalized roof-trace measure

$$d\nu_{\omega}^{(0)} := \frac{d\mu_{\omega}^{(0)}}{Z_{\omega}^{(0)}}. \quad (\text{S91})$$

The branch-reweighted auxiliary measure at depth ϵ is

$$d\mu_{\omega,\epsilon}^{\text{bl}} = \rho_{\omega,\epsilon}^{\text{bl}}(\boldsymbol{\xi}, t) d^2 \boldsymbol{\xi} dt = b_{\omega,\epsilon}(\boldsymbol{\xi}, t) d\mu_{\omega}^{(0)}. \quad (\text{S92})$$

After normalization,

$$d\nu_{\omega,\epsilon}^{\text{bl}} = \frac{d\mu_{\omega,\epsilon}^{\text{bl}}}{\int d\mu_{\omega,\epsilon}^{\text{bl}}} = \frac{b_{\omega,\epsilon}(\boldsymbol{\xi}, t)}{\mathbb{E}_{\nu_{\omega}^{(0)}}[b_{\omega,\epsilon}]} d\nu_{\omega}^{(0)}. \quad (\text{S93})$$

This identity is exact for the homogeneous branch probe. Now for any finite-window observable $F(\boldsymbol{\xi}, t)$, its expectation with respect to this auxiliary measure is therefore

$$\mathbb{E}_{\nu_{\omega,\epsilon}^{\text{bl}}}[F] = \frac{\mathbb{E}_{\nu_{\omega}^{(0)}}[F b_{\omega,\epsilon}]}{\mathbb{E}_{\nu_{\omega}^{(0)}}[b_{\omega,\epsilon}]}. \quad (\text{S94})$$

Differentiating Eq. (S94) at $\epsilon = 0$, and using Eq. (S89) gives

$$\partial_\epsilon \mathbb{E}_{\nu_{\omega,\epsilon}^{\text{bl}}} [F] \Big|_{\epsilon=0} = \mathbb{E}_{\nu_\omega^{(0)}} [F Ra_\omega] - \mathbb{E}_{\nu_\omega^{(0)}} [F] \mathbb{E}_{\nu_\omega^{(0)}} [Ra_\omega]. \quad (\text{S95})$$

Thus

$$\partial_\epsilon \mathbb{E}_{\nu_{\omega,\epsilon}^{\text{bl}}} [F] \Big|_{\epsilon=0} = \text{Cov}_{\nu_\omega^{(0)}} (F, Ra_\omega). \quad (\text{S96})$$

Taking $F = t$ to calculate the normalized first time moment, the local first-order boundary-response coefficient is

$$\Lambda_\omega := \partial_\epsilon \bar{t}_{\text{bl}}(\omega, \epsilon) \Big|_{\epsilon=0}, \quad \bar{t}_{\text{bl}}(\omega, \epsilon) := \mathbb{E}_{\nu_{\omega,\epsilon}^{\text{bl}}} [t], \quad (\text{S97})$$

and therefore

$$\Lambda_\omega = \text{Cov}_{\nu_\omega^{(0)}} (t, Ra_\omega). \quad (\text{S98})$$

This is the desired covariance form of the local boundary response. The derivation used the exact branch exponential through $b_{\omega,\epsilon}$, and the only derivative taken was the derivative at the physical roof, $\epsilon = 0$. For the harmonic transverse family we write

$$R = \sqrt{\omega} s, \quad (\text{S99})$$

where s is the dimensionless tangential radius in oscillator units. Hence

$$\Lambda_\omega = \sqrt{\omega} \text{Cov}_{\nu_\omega^{(0)}} (t, sa_\omega) = \sqrt{\omega} \beta_\omega^{\text{bl}}, \quad (\text{S100})$$

with

$$\beta_\omega^{\text{bl}} := \text{Cov}_{\nu_\omega^{(0)}} (t, sa_\omega). \quad (\text{S101})$$

Since $|a_\omega| \leq 2$ and $0 \leq t \leq T$ on the finite observation window, Cauchy–Schwarz gives, for example,

$$|\beta_\omega^{\text{bl}}| \leq 2T \sqrt{\mathbb{E}_{\nu_\omega^{(0)}} [s^2]}. \quad (\text{S102})$$

Thus β_ω^{bl} remains bounded whenever the scaled tangential moment $\mathbb{E}_{\nu_\omega^{(0)}} [s^2]$ remains order one, as checked numerically in Sec. S7.

We emphasize that $\bar{t}_{\text{bl}}(\omega, \epsilon)$ is not an observable detection time at $z = L - \epsilon$; all physical detection statistics are defined by the roof flux at $z = L$. The auxiliary boundary-response calculation only extracts the local source in Eq. S100. In the full finite guide, this local filtering affects the roof statistics through the first near-roof interaction and through undetected, branch-reweighted components that leave the near-roof region and later return, as represented by the reduced flux closure in Sec. S8.

S7. FINITE-GRID BOUNDARY-SYMBOL DIAGNOSTIC AND DUHAMEL ESTIMATE

This section gives the discrete diagnostic corresponding to Sec. S6. The physical detector statistic is always the roof flux at $z = L$. The auxiliary depth

$$\epsilon_g = h_z = \frac{L}{N_z} \quad (\text{S103})$$

is used only as a one-grid normal probe of the spinor-ABC boundary response. It is not a physical detector thickness. At each diagnostic time we use the stored roof trace

$$W_\omega(\boldsymbol{\xi}, t) = \widehat{\Psi}(\boldsymbol{\xi}, L, t). \quad (\text{S104})$$

To match the central finite-difference tangential derivatives used in the solver, the continuum tangential variable is replaced by the discrete symbol

$$\boldsymbol{\xi}_h(k) = \left(\frac{\sin(k_x h_x)}{h_x}, \frac{\sin(k_y h_y)}{h_y} \right), \quad R_h = |\boldsymbol{\xi}_h|, \quad s = \frac{R_h}{\sqrt{\omega}}. \quad (\text{S105})$$

Because the transverse walls are Dirichlet, this symbol is used as a local stencil diagnostic for the central-difference tangential derivative, not as an exact diagonalization of the full finite transverse operator. The discrete spinor generator $\mathcal{J}_h$ is obtained from Eq. (S68) by replacing $\boldsymbol{\xi}$ with $\boldsymbol{\xi}_h$, so that $\mathcal{J}_h^2 = R_h^2 I$. For $R_h > 0$, define

$$\widehat{\mathcal{J}}_h = \frac{\mathcal{J}_h}{R_h}. \quad (\text{S106})$$

The local spinor factor is

$$a_\omega(\boldsymbol{\xi}, t) = -2 \frac{W_\omega^\dagger \widehat{\mathcal{J}}_h W_\omega}{W_\omega^\dagger W_\omega}, \quad |a_\omega| \leq 2. \quad (\text{S107})$$

All averages are taken with respect to the normalized roof-trace measure

$$\mathbb{E}_{\nu_\omega^{(0)}}[F] = \frac{\int_0^T \sum_{\boldsymbol{\xi}} \|W_\omega(\boldsymbol{\xi}, t)\|^2 F(\boldsymbol{\xi}, t) dt}{\int_0^T \sum_{\boldsymbol{\xi}} \|W_\omega(\boldsymbol{\xi}, t)\|^2 dt}. \quad (\text{S108})$$

We compute

$$\beta_\omega^{\text{bl}} = \text{Cov}_{\nu_\omega^{(0)}}(t, s a_\omega), \quad \Lambda_\omega = \text{Cov}_{\nu_\omega^{(0)}}(t, R_h a_\omega) = \sqrt{\omega} \beta_\omega^{\text{bl}}. \quad (\text{S109})$$

We also monitor the one-grid finite-depth probe scale

$$q_\omega = \mathbb{E}_{\nu_\omega^{(0)}}[(\epsilon_g R_h)^2] = (\epsilon_g \sqrt{\omega})^2 \mathbb{E}_{\nu_\omega^{(0)}}[s^2]. \quad (\text{S110})$$

The quantity q_ω is not used in the derivation of the covariance formula in Sec. S6.4. It is only a dimensionless finite-depth diagnostic for the one-grid probe $\epsilon_g = h_z$: small q_ω means that the sampled

modes have $\epsilon_g R_h$ small on average. It is not an error estimate. The actual one-grid continuation defect is monitored separately by $\mathcal{E}_{2\text{br}}$. As a global consistency check, we record the detector-budget mismatch

$$\Delta_{\text{bud}}^{\text{rel}} = \left| \frac{\int_0^T J_L(t) dt - (\|\Psi_0\|^2 - \|\Psi_T\|^2)}{\|\Psi_0\|^2 - \|\Psi_T\|^2} \right|. \quad (\text{S111})$$

Finally, we compare the true trace one grid spacing below the roof with the homogeneous two-branch continuation from the roof trace:

$$W_{\text{hom}}(\boldsymbol{\xi}, t; h_z) = e^{-i\kappa h_z} e^{-h_z \mathcal{J}_h(\boldsymbol{\xi})} W_\omega(\boldsymbol{\xi}, t), \quad W_{\text{in}}(\boldsymbol{\xi}, t) = \widehat{\Psi}(\boldsymbol{\xi}, L - h_z, t). \quad (\text{S112})$$

The one-grid continuation defect is

$$\mathcal{E}_{2\text{br}}(h_z) = \left\langle \frac{\|W_{\text{in}}(\cdot, t) - W_{\text{hom}}(\cdot, t; h_z)\|_{\ell_\xi^2}}{\|W_{\text{in}}(\cdot, t)\|_{\ell_\xi^2}} \right\rangle_{\text{act}}. \quad (\text{S113})$$

Here $\langle \cdot \rangle_{\text{act}}$ denotes an active-flux-weighted average over times for which the roof signal is appreciable.

TABLE S4. Representative one-grid diagnostics for the spinor-ABC boundary response. Here $\epsilon_g = h_z$, $R_h = |\boldsymbol{\xi}_h|$, $s = R_h/\sqrt{\omega}$, and $q_\omega = \mathbb{E}[(\epsilon_g R_h)^2]$ is a one-grid finite-depth probe scale, not an error estimate and not an assumption in the covariance derivation.

ω	$\epsilon_g \sqrt{\omega}$	q_ω	$\Delta_{\text{bud}}^{\text{rel}}$	$\mathbb{E}[s^2]$	β_ω^{bl}	Λ_ω	$\mathcal{E}_{2\text{br}}$
1	0.013	1.7×10^{-4}	1.8×10^{-5}	0.978	-8.0×10^{-2}	-8.0×10^{-2}	1.8×10^{-3}
50	0.094	2.4×10^{-2}	4.0×10^{-4}	2.744	-2.5×10^{-2}	-1.8×10^{-1}	1.5×10^{-2}
100	0.133	5.6×10^{-2}	7.2×10^{-4}	3.140	3.2×10^{-2}	3.2×10^{-1}	3.0×10^{-2}
200	0.189	1.0×10^{-1}	6.8×10^{-4}	2.919	2.1×10^{-2}	3.0×10^{-1}	5.4×10^{-2}
300	0.231	1.4×10^{-1}	2.4×10^{-4}	2.650	1.0×10^{-2}	1.8×10^{-1}	7.6×10^{-2}

The detector-budget mismatch stays below 10^{-3} . The scaled tangential moment $\mathbb{E}[s^2]$ remains order-one, confirming that the relevant discrete tangential scale is $R_h = \sqrt{\omega} s$. The covariance coefficient β_ω^{bl} remains bounded, so the local source has the form $\Lambda_\omega = \sqrt{\omega} \beta_\omega^{\text{bl}}$. The sign of β_ω^{bl} is not itself the mechanism; the diagnostic supports the scale $R_h \sim \sqrt{\omega}$. The two-branch continuation defect remains below about 8%, showing that the homogeneous spinor-ABC continuation captures the near-roof trace at one grid depth with controlled finite-grid defect.

S7.1. Duhamel estimate

This subsection proves the estimate used in Sec. S6.1. Starting from Eq. (S64), assume that $\widehat{\mathcal{E}}_{\text{bc}}$ is C^1 in the normal variable on $L - \epsilon \leq z \leq L$. Define

$$G_\omega(\boldsymbol{\xi}, t; \epsilon) = \sup_{0 \leq u \leq \epsilon} \left\| \partial_z \widehat{\mathcal{E}}_{\text{bc}}(\boldsymbol{\xi}, L - u, t) \right\|. \quad (\text{S114})$$

Since $\widehat{\mathcal{E}}_{\text{bc}}(\boldsymbol{\xi}, L, t) = 0$,

$$\left\| \widehat{\mathcal{E}}_{\text{bc}}(\boldsymbol{\xi}, L - u, t) \right\| \leq u G_{\omega}(\boldsymbol{\xi}, t; \epsilon). \quad (\text{S115})$$

Moreover, from $\mathcal{C}(\boldsymbol{\xi}) = i\kappa I + R\Gamma(\boldsymbol{\xi})$, $\Gamma^2 = I$, and $R = |\boldsymbol{\xi}|$,

$$\left\| e^{-(\epsilon-u)\mathcal{C}(\boldsymbol{\xi})} \right\| \leq e^{(\epsilon-u)R} \leq e^{\epsilon R}. \quad (\text{S116})$$

Inserting Eqs. (S115) and (S116) into Eq. (S64) gives

$$\begin{aligned} \left\| \widehat{\mathcal{E}}_{\text{Duh}}(\boldsymbol{\xi}, L - \epsilon, t) \right\| &\leq e^{\epsilon R} G_{\omega}(\boldsymbol{\xi}, t; \epsilon) \int_0^{\epsilon} u \, du \\ &= \frac{\epsilon^2}{2} e^{\epsilon R} G_{\omega}(\boldsymbol{\xi}, t; \epsilon). \end{aligned} \quad (\text{S117})$$

Thus, for each fixed tangential mode,

$$\widehat{\mathcal{E}}_{\text{Duh}}(\boldsymbol{\xi}, L - \epsilon, t) = O(\epsilon^2) \quad (\epsilon \rightarrow 0). \quad (\text{S118})$$

In particular,

$$\widehat{\mathcal{E}}_{\text{Duh}}(\boldsymbol{\xi}, L, t) = 0, \quad \partial_{\epsilon} \widehat{\mathcal{E}}_{\text{Duh}}(\boldsymbol{\xi}, L - \epsilon, t) \Big|_{\epsilon=0} = 0. \quad (\text{S119})$$

Therefore the Duhamel remainder does not change the first inward normal variation of the roof trace.

The finite-grid counterpart of this statement is the continuation defect reported in Table S4.

The estimate S118–S119 is modewise; no uniform continuum bound in $\boldsymbol{\xi}$ or ω is claimed here. In the finite-grid implementation the sampled tangential spectrum is finite, and the size of the one-grid probe is diagnosed by q_{ω} , while the actual continuation mismatch is measured by $\mathcal{E}_{2\text{br}}$ in Table S4.

S8. REDUCED RETURN CLOSURE

The boundary-symbol calculation identifies the local spinor-ABC scale

$$\Lambda_{\omega} = \sqrt{\omega} \beta_{\omega}^{\text{bl}}. \quad (\text{S120})$$

This is not yet a detection-time statistic. The physical observable is the detector-present roof flux

$$g(t; \omega) = J_L(t). \quad (\text{S121})$$

The purpose of this section is to give a reduced flux-balance and finite-guide memory closure explaining how undetected spinor-ABC-filtered components can return to the roof and make the local scale $R = |\boldsymbol{\xi}| \sim \sqrt{\omega}$ visible in finite-window moments.

Introduce a near-roof matching plane and an auxiliary control layer

$$\Sigma_* = \{z = z_*\}, \quad z_* = L - \ell, \quad \Omega_{\text{bl}}^{(\ell)} = \{(x, y, z) : z_* < z < L\}. \quad (\text{S122})$$

Let

$$P_{\text{bl}}^{(\ell)}(t) = \int_{\Omega_{\text{bl}}^{(\ell)}} \rho(x, y, z, t) dV, \quad (\text{S123})$$

and define the signed matching-plane flux and the detected roof flux by

$$J_*(t) = \int_{\Sigma_*} j_z(x, y, z_*, t) dxdy, \quad J_L(t) = \int_{\Sigma_L} j_z(x, y, L, t) dxdy. \quad (\text{S124})$$

Here j_z denotes the z -component of the Pauli current. The sign convention is that $J_*(t) > 0$ means flux entering the near-roof layer from below. Since the spinor ABC gives

$$J_L(t) = \kappa \int_{\Sigma_L} \rho(x, y, L, t) dxdy \geq 0, \quad (\text{S125})$$

the continuity equation in the auxiliary layer gives

$$\frac{d}{dt} P_{\text{bl}}^{(\ell)}(t) = J_*(t) - J_L(t). \quad (\text{S126})$$

Thus probability entering the near-roof layer may be detected at the roof, remain temporarily stored in the layer, or leave the layer again through Σ_* as an undetected filtered component.

Let $F_{\text{out}}^{\text{nc}}(t; \omega, \ell)$ denote the effective outgoing no-click flux of undetected, spinor-ABC-filtered amplitude through Σ_* . This object is not measured directly and is not equal to the detected roof flux. A causal return kernel $k_{\omega, \ell}^{\text{ret}}(u)$ then represents the later contribution to the roof flux from such undetected components after propagation through the finite guide. Here and below $u \geq 0$ denotes a time lag: $F_{\text{out}}^{\text{nc}}(t - u; \omega, \ell)$ is the no-click flux that left the near-roof layer at the earlier time $t - u$, and $k_{\omega, \ell}^{\text{ret}}(u)$ is the effective weight for that component to contribute to the roof flux after delay u . Therefore

$$g_{\text{box}}(t; \omega) = g_{\text{first}}(t; \omega) + \int_0^t k_{\omega, \ell}^{\text{ret}}(u) F_{\text{out}}^{\text{nc}}(t - u; \omega, \ell) du, \quad (\text{S127})$$

where g_{first} denotes the roof flux associated with the first near-roof encounter before a full return from the guide. Equation (S127) is a reduced flux-level closure, not an exact mode-resolved TDSE decomposition. It summarizes the delayed influence of an unresolved no-click channel after phase, branch, and mode information have been compressed into an effective scalar kernel.

For a one-channel moment estimate, one may eliminate the unresolved no-click flux $F_{\text{out}}^{\text{nc}}$ into an effective causal memory kernel $k_{\omega, \ell}^{\text{eff}}$. This second kernel should not be identified with $k_{\omega, \ell}^{\text{ret}}$: it also contains the unresolved relation between the hidden no-click flux and the previous roof/near-roof history. This gives the scalar Volterra closure

$$g_{\text{box}} = g_{\text{first}} + K_{\omega, \ell}^{\text{eff}} g_{\text{box}}, \quad (K_{\omega, \ell}^{\text{eff}} f)(t) = \int_0^t k_{\omega, \ell}^{\text{eff}}(u) f(t - u) du. \quad (\text{S128})$$

The occurrence of g_{box} on the right-hand side should not be read as already detected probability returning to the detector. It is the result of eliminating the unresolved undetected channel into an effective memory kernel. The effective kernel describes the delayed influence of surviving, branch-reweighted amplitude.

If $K_{\omega,\ell}^{\text{eff}}$ is contractive, for example $\|k_{\omega,\ell}^{\text{eff}}\|_1 < 1$, then formally

$$g_{\text{box}} = (I - K_{\omega,\ell}^{\text{eff}})^{-1} g_{\text{first}} = \sum_{n=0}^{\infty} (K_{\omega,\ell}^{\text{eff}})^n g_{\text{first}}. \quad (\text{S129})$$

For an untruncated causal flux f , define

$$M_0[f] = \int_0^{\infty} f(t) dt, \quad M_1[f] = \int_0^{\infty} t f(t) dt. \quad (\text{S130})$$

Let

$$m_{\text{ret}}^{(0)} = \int_0^{\infty} k_{\omega,\ell}^{\text{eff}}(u) du, \quad m_{\text{ret}}^{(1)} = \int_0^{\infty} u k_{\omega,\ell}^{\text{eff}}(u) du, \quad (\text{S131})$$

when $m_{\text{ret}}^{(0)} \neq 0$, define the normalized first moment of the effective returned channel by

$$\bar{U}_{\text{ret}}(\omega; \ell) := \frac{m_{\text{ret}}^{(1)}}{m_{\text{ret}}^{(0)}}. \quad (\text{S132})$$

If $k_{\omega,\ell}^{\text{eff}}$ is nonnegative, $m_{\text{ret}}^{(0)}$ may be interpreted as an effective return probability and $\bar{U}_{\text{ret}}$ as the mean return delay. In general, they are only the zeroth moment and normalized first moment of the reduced memory kernel. The corresponding convolution identities therefore yield

$$M_0[K_{\omega,\ell}^{\text{eff}} f] = m_{\text{ret}}^{(0)} M_0[f], \quad (\text{S133})$$

and

$$M_1[K_{\omega,\ell}^{\text{eff}} f] = m_{\text{ret}}^{(0)} M_1[f] + m_{\text{ret}}^{(1)} M_0[f]. \quad (\text{S134})$$

Consequently, for the corresponding untruncated mean time,

$$\bar{\tau}_{\text{box}} = \bar{\tau}_{\text{first}} + \frac{m_{\text{ret}}^{(1)}}{1 - m_{\text{ret}}^{(0)}}. \quad (\text{S135})$$

This equation is not used as a fit model for the numerical data. It only shows how a delayed effective memory kernel shifts a flux first moment.

The local spinor-ABC filtering enters the returned component through the same tangential scale inherited from the roof response. For the harmonic transverse family,

$$R = |\boldsymbol{\xi}| = \sqrt{\omega} s, \quad (\text{S136})$$

with s dimensionless and order one. The returned component has been filtered by the boundary, propagated through the finite guide, and re-entered the near-roof region. Therefore its effective coefficient need not equal $\beta_{\omega}^{\text{bl}}$. We write this returned-channel coefficient as $\beta_{\omega}^{\text{ret}}$, and use the reduced moment ansatz

$$\bar{U}_{\text{ret}}(\omega; \ell) := \frac{m_{\text{ret}}^{(1)}}{m_{\text{ret}}^{(0)}} \simeq U_{\text{bulk,ret}}(\ell) + c_0(\ell) + c_1(\ell) \sqrt{\omega} \beta_{\omega}^{\text{ret}}. \quad (\text{S137})$$

Since $m_{\text{ret}}^{(1)} = m_{\text{ret}}^{(0)} \bar{U}_{\text{ret}}$, Eq. (S135) can be written as

$$\bar{\tau}_{\text{box}} = \bar{\tau}_{\text{first}} + \alpha_{\text{ret}}(\omega; \ell) \bar{U}_{\text{ret}}(\omega; \ell), \quad \alpha_{\text{ret}}(\omega; \ell) := \frac{m_{\text{ret}}^{(0)}(\omega; \ell)}{1 - m_{\text{ret}}^{(0)}(\omega; \ell)}. \quad (\text{S138})$$

Thus $m_{\text{ret}}^{(0)}$ controls the effective amount of returned weight, while $\bar{U}_{\text{ret}}$ controls its effective delay. Combining Eqs. (S135) and (S137) gives, over a finite fitted range,

$$\bar{\tau}_{\text{box}}(\omega) \simeq A_0(\ell, T) + B_0(\ell, T) \sqrt{\omega} + \mathcal{R}_{\text{eff}}(\omega; \ell, T), \quad (\text{S139})$$

where $\mathcal{R}_{\text{eff}}$ collects finite-window truncation, the variation of the effective return probability, and the residual ω -dependence of the returned-channel coefficient. When this remainder varies slowly over the sampled range, the finite-window moment is well approximated by an affine function of $\sqrt{\omega}$.

Finally, the detection statistic plotted in the main text is the finite-window restricted mean detection-time statistics, not the untruncated conditional mean. As shown in Sec. S4,

$$\mu^*(T; \omega) = D_T(\omega) \bar{\tau}_T^{\text{cond}}(\omega) + [1 - D_T(\omega)]T. \quad (\text{S140})$$

Therefore the observed finite-window coefficient receives contributions both from delayed conditional timing and from the decreasing detected fraction.

Importantly, the closure is not a mode-resolved solution of the TDSE and is not used to fit the data. Its only purpose is to show how an undetected no-click channel, once compressed into an effective causal memory kernel, can transfer the local boundary scale $R \sim \sqrt{\omega}$ into finite-window roof-flux moments. Therefore the observed affine-in- $\sqrt{\omega}$ behavior should be read as a finite-range detector-response signature, not as a universal asymptotic law.

S9. CONTROL CHECKS

This section records exploratory control checks addressing two questions: whether the spinor-ABC delayed sector is removed by representative bulk spin perturbations when the spinor ABC is retained, and whether the same confinement trend can be reproduced without the spinor ABC by using generic spin-dependent absorbing Hamiltonian layers. These controls are not used to establish the fitted scaling or to prove uniqueness of the spinor ABC. Within the tested parameter ranges, bulk spin perturbations did not substantially remove the delayed sector, and generic absorbing Hamiltonian layers did not robustly reproduce the spinor-ABC confinement trend.

S9.1. Bulk spin and transverse-state controls

We first kept the spinor absorbing boundary fixed and modified only the bulk Hamiltonian,

$$H = H_{\text{waveguide}}(\omega) + H_{\text{spin}}. \quad (\text{S141})$$

Representative choices of H_{spin} included localized Rabi–Zeeman terms, transverse Zeeman gradients, Rashba and Rashba–Dresselhaus spin–orbit couplings, and a Pauli magnetic minimal-coupling Hamiltonian. In the magnetic case, the roof condition was replaced by the corresponding gauge-covariant spinor ABC,

$$(\boldsymbol{\sigma} \cdot \boldsymbol{D})\Psi = i\kappa\sigma_z\Psi, \quad \boldsymbol{D} = \boldsymbol{\nabla} - iq\boldsymbol{A}. \quad (\text{S142})$$

We also tested whether the effect depends essentially on preparing the transverse harmonic ground state by replacing $\chi_{00}^{(\omega)}$ with first excited transverse modes and with the chiral combination

$$\chi_{\text{chiral}}^{(\omega)} = \frac{\chi_{10}^{(\omega)} + i\chi_{01}^{(\omega)}}{\sqrt{2}}. \quad (\text{S143})$$

Within the tested parameter range, these modifications did not substantially remove the delayed sector relative to the spinor-ABC reference run. Thus the confinement dependence is not primarily caused by generic bulk spin dynamics or by the special choice of the transverse ground state. The dominant source remains the boundary response.

S9.2. Hamiltonian-layer mimics

We next asked whether the spinor-ABC delay can be reproduced without imposing the spinor ABC at the detector surface. We replaced the boundary condition by a finite absorbing Hamiltonian layer,

$$L_{\text{det}} < z < L_{\text{det}} + \ell_{\text{det}}, \quad (\text{S144})$$

with a smooth turn-on function $\chi(z)$ and a Dirichlet wall at the far end of the layer:

$$H = H_{\text{waveguide}}(\omega) + H_{\text{layer}}. \quad (\text{S145})$$

We considered ordinary spin-dependent absorbing layers, including Rashba-type loss and helicity-based tangential-momentum loss. These models did not robustly reproduce the spinor-ABC confinement dependence. For comparison, we also considered an inverse-engineered impedance layer. Let

$$K = \sigma_x p_y - \sigma_y p_x, \quad R = (p_x^2 + p_y^2)^{1/2}, \quad p_j = -i\partial_j. \quad (\text{S146})$$

In tangential Fourier variables, K has eigenvalues $\pm R$, and K/R separates the two tangential spin-momentum branches. The inverse-engineered layer was chosen so that, in a stationary Dirichlet-backed calibration, its two branch impedances approximate

$$i\kappa - R, \quad i\kappa + R. \quad (\text{S147})$$

This model is not a generic absorbing layer. It explicitly builds the spinor-ABC branch impedances into the layer. Therefore these controls support a more specific conclusion: generic absorbing layers or scalar absorber reflection do not reproduce the spinor-ABC confinement trend. Separately, the bulk perturbation checks in Sec. S9.1 show that representative bulk spin dynamics do not explain away the delayed sector when the spinor ABC is retained. Reproducing the response requires an effective entrance impedance with the same tangential branch structure $i\kappa \pm |\xi|$ by construction.

S10. PARAMETER SCANS AND ROBUSTNESS CHECKS

This section records how the spinor-ABC roof-flux statistics change when source or detector parameters are varied. These scans are not used to derive a new scaling law. Their purpose is to show that fitted coefficients, detected fractions, and late-time tails depend on the finite-window source/detector geometry, while the local transverse scale remains

$$R = |\xi| \sim \sqrt{\omega}. \quad (\text{S148})$$

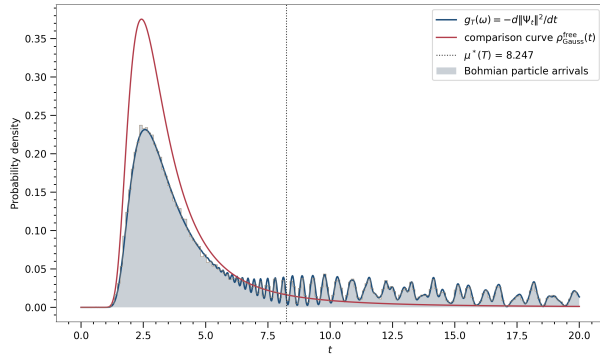
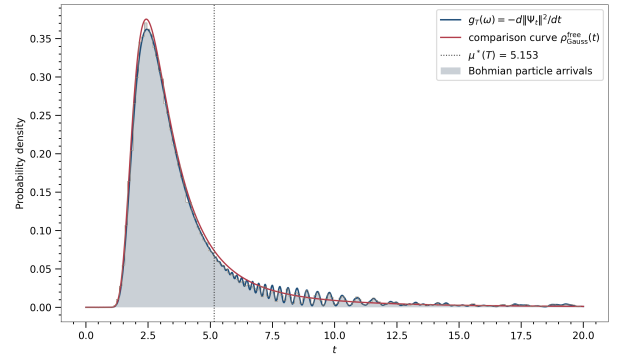
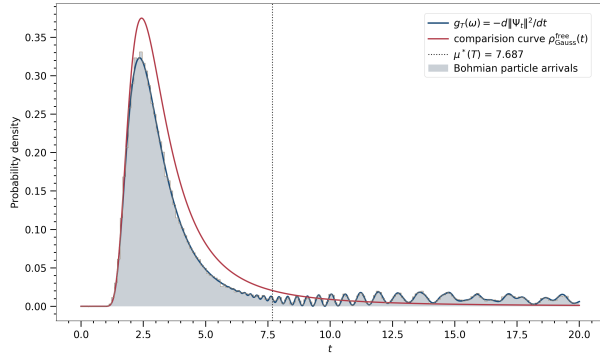
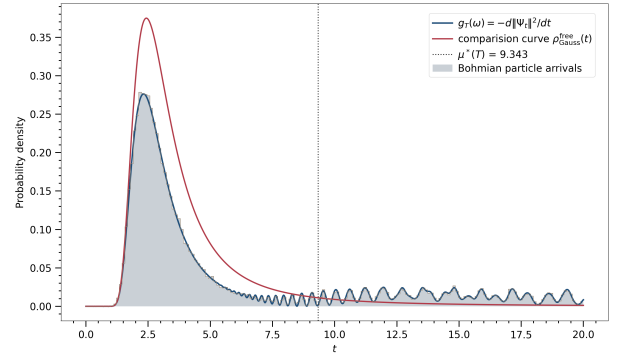
Thus, when an affine-in- $\sqrt{\omega}$ fit is used, its coefficients should be read as source-, detector-, geometry-, and window-dependent.

S10.1. κ -scan

Figure S2 and Table S5 show the dependence on the boundary parameter κ , with $k_z = \pi$, $\sigma_{\parallel} = 0.5$, and $\omega = 1$ fixed. The finite-window statistic is smallest near the longitudinally matched value $\kappa \simeq k_z = \pi$. Moving κ below or above this value increases $\mu^*(T; \kappa)$ and changes the late-time tail.

This is consistent with the scalar one-dimensional reference picture, in which a Robin absorber is best matched near $\kappa = k_z$. For the spinor ABC this is only a longitudinal matching guide: the transverse confinement mechanism remains controlled by

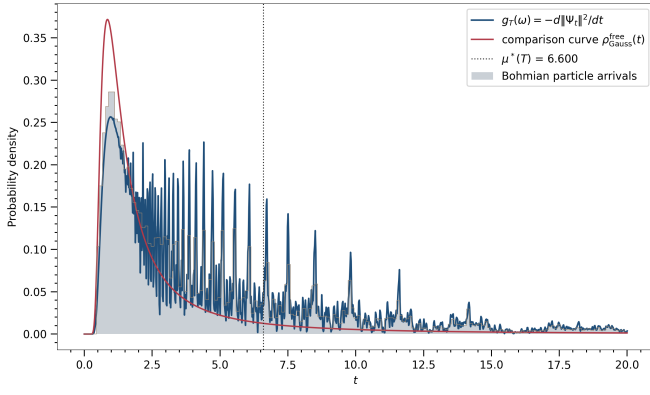
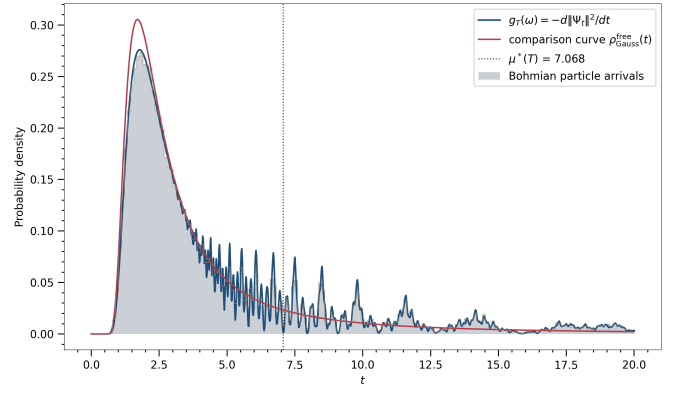
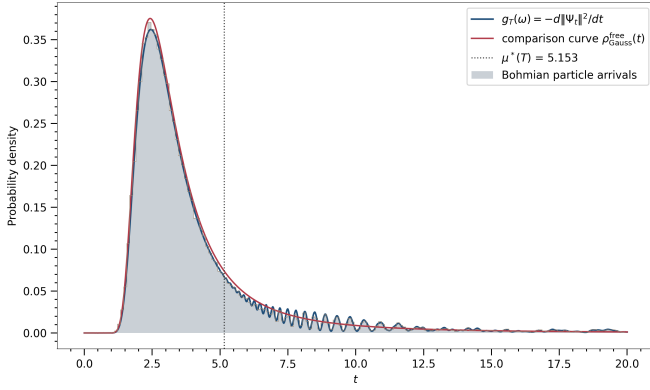
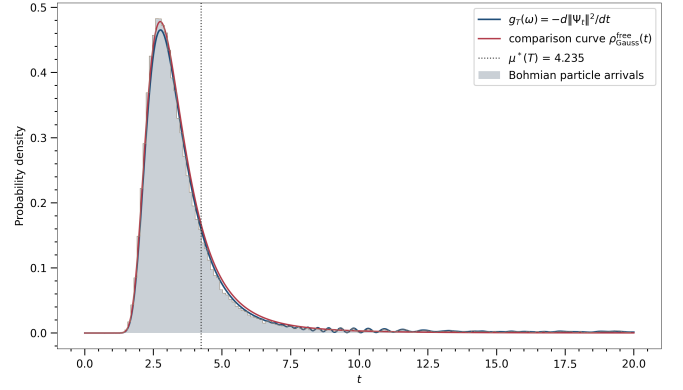
$$R = |\xi| \sim \sqrt{\omega}. \quad (\text{S149})$$

(a) $\kappa = 1$ (b) $\kappa = \pi$ (c) $\kappa = 9$ (d) $\kappa = 13$ FIG. S2. κ -scan for the spinor ABC. The parameters $k_z = \pi$, $\sigma_{\parallel} = 0.5$, and $\omega = 1$ are fixed.TABLE S5. Finite-window statistic $\mu^*(T; \kappa)$ for the κ -scan at fixed $\omega = 1$, $k_z = \pi$, and $\sigma_{\parallel} = 0.5$.

κ	0.1	0.5	1	2	π	5	7	9	13
$\mu^*(T; \kappa)$	17.016	11.709	8.247	5.653	5.153	5.729	6.717	7.687	9.343

S10.2. $\sigma_{\parallel}$ -scan

Changing $\sigma_{\parallel}$ changes the longitudinal spatial width and momentum spread of the incoming packet. Narrower packets have broader momentum support and produce broader, more oscillatory detector-time profiles. Therefore $\sigma_{\parallel}$ affects the finite-window coefficients and detailed tails, but not the local origin of the confinement scale.

(a) $\sigma_{\parallel} = 0.1$ (b) $\sigma_{\parallel} = 0.25$ (c) $\sigma_{\parallel} = 0.5$ (d) $\sigma_{\parallel} = 0.75$ FIG. S3. $\sigma_{\parallel}$ -scan for the spinor ABC. The parameters $k_z = \kappa = \pi$ and $\omega = 1$ are fixed.

For two representative longitudinal widths, the finite-window fits are

$$\mu^*(T; \omega) \simeq A_{\sigma} + B_{\sigma} \sqrt{\omega}. \quad (\text{S150})$$

TABLE S6. Representative fit coefficients for $\mu^*(T; \omega) \simeq A_{\sigma} + B_{\sigma} \sqrt{\omega}$, with $k_z = \kappa = \pi$ fixed.

$\sigma_{\parallel}$	0.25	0.5
A_{σ}	6.253	4.084
B_{σ}	0.435	0.638

The coefficients change with the source packet, but the approximate $\sqrt{\omega}$ trend persists over the finite fitted range.

S10.3. k_z -scan

Changing k_z mainly changes the bulk flight-time scale and the longitudinal matching to the detector. For small k_z , the incoming packet is slow and the fixed boundary parameter $\kappa = \pi$ is strongly mismatched;

the roof-flux density is broad and the late oscillatory sector is prominent. For larger k_z , the first roof encounter occurs earlier and the finite-window statistic decreases. Thus k_z changes effective prefactors and tail structure, while the transverse-confinement mechanism remains the boundary scale $R = |\xi| \sim \sqrt{\omega}$.

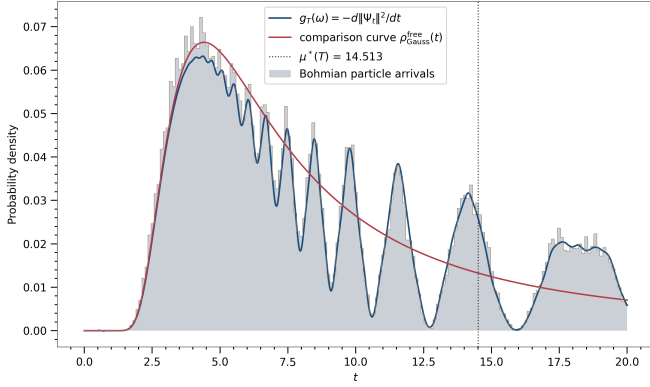
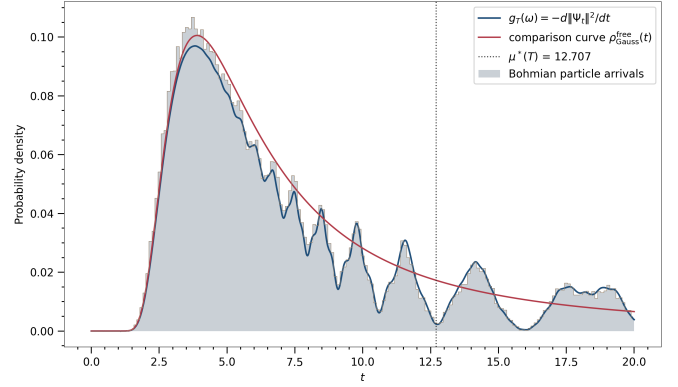
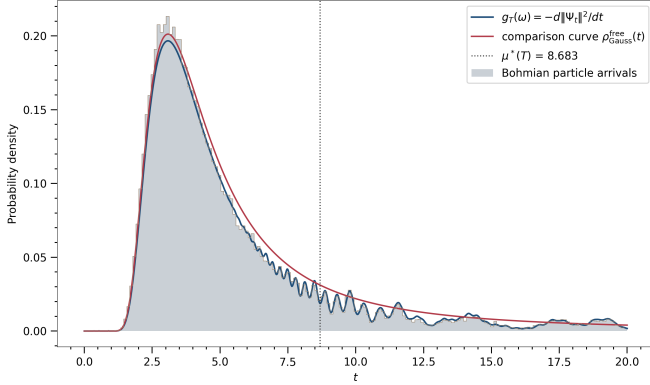
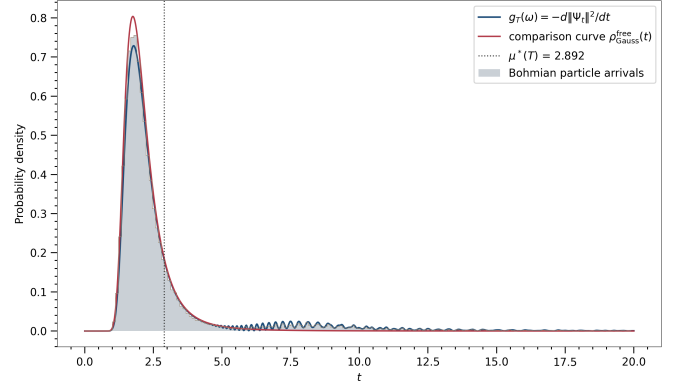
(a) $k_z = 0.5$ (b) $k_z = 1$ (c) $k_z = 2$ (d) $k_z = 5$

FIG. S4. k_z -scan for the spinor ABC. The parameters $\kappa = \pi$, $\sigma_{\parallel} = 0.5$, and $\omega = 1$ are fixed.

S10.4. Large-box check

We repeated representative spinor-ABC runs in a longer box, $L_z = 100$, with $z_c = 50$, $k_z = \kappa = \pi$, and $\sigma_{\parallel} = 0.5$. Increasing ω again suppresses the prompt roof-flux peak and enhances a delayed oscillatory sector. Thus the confinement-induced delayed sector is not only a short-box artifact.

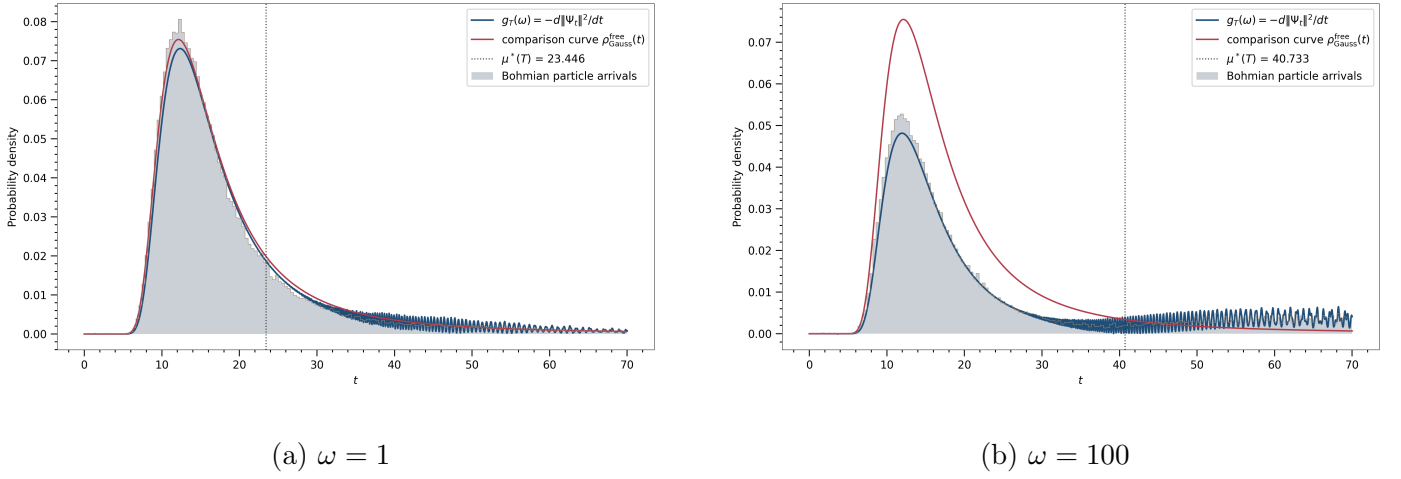


FIG. S5. Large-box spinor-ABC check. Only ω is varied; the delayed sector persists.

Overall, the scans show that source and detector parameters change the numerical coefficients, detected fractions, and detailed oscillatory tails. They do not alter the local origin of the confinement dependence:

$$R = |\xi| \sim \sqrt{\omega}. \quad (\text{S151})$$

S11. PHYSICAL SCALES

The analysis above is dimensionless. Let a_0 be the physical length represented by one dimensionless unit. Then

$$r_{\text{phys}} = a_0 r, \quad t_{\text{phys}} = T_* t, \quad T_* = \frac{m_{\text{phys}} a_0^2}{\hbar}. \quad (\text{S152})$$

The transverse trap frequency and longitudinal wave numbers scale as

$$\Omega_{\perp} = \frac{\omega}{T_*}, \quad k_{z,\text{phys}} = \frac{k_z}{a_0}, \quad \kappa_{\text{phys}} = \frac{\kappa}{a_0}. \quad (\text{S153})$$

The physical transverse oscillator length is

$$\ell_{\perp,\text{phys}} = \frac{a_0}{\sqrt{\omega}}. \quad (\text{S154})$$

As an illustrative scale, for single ^{87}Rb atoms and $a_0 = 2 \mu\text{m}$, one obtains

$$T_* \simeq 5.47 \text{ ms}. \quad (\text{S155})$$

Thus the observation window $T = 20$ corresponds to approximately 109 ms. The values $\omega = 1, 100, 300$ correspond to transverse oscillator lengths approximately $2.0 \mu\text{m}$, $0.20 \mu\text{m}$, and $0.115 \mu\text{m}$, respectively.

These numbers are reference scales only; the main text does not propose a complete detector design. The experimentally relevant signature is the confinement-controlled deformation of the detector-present

roof-flux distribution: increasing ω suppresses the early peak, reduces the detected fraction in a fixed observation window, and enhances the delayed oscillatory sector. Over the pre-saturation regime studied here, this enters the finite-window statistic through the local scale $\sqrt{\omega}$.

A literal implementation of the spinor ABC would require an effective absorbing boundary whose entrance response contains the tangential spin–momentum branches

$$i\kappa \pm |\boldsymbol{\xi}|. \tag{S156}$$

Generic spin-dependent loss or bulk spin–orbit terms should not be expected to reproduce this response without engineering the corresponding boundary impedance. The inverse-engineered impedance layer in Sec. [S9.2](#) should be viewed only as a proof-of-principle design target: it approaches the spinor-ABC response precisely because the two branch impedances are inserted by construction.